



Description of the degree program

# Digital Communication and Media Technologies (Master)

## PO 4

Date: 01.10.2025

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### Master Digital Communication and Media Technology

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------|------------------------------|
| Interdisciplinary Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |                   | 5 ECTS                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |                   |                              |
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Interdisciplinary Module |                   |                              |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2199130                  | Module version    |                              |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MW-STD-13                | Language          |                              |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | only in the winter term  | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 Semester               | Institution       |                              |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2 / 5,0                  | Module owner      |                              |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |                   |                              |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 30                       | Self studying (h) | 120                          |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |                   |                              |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                          |                   |                              |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1 presentation           |                   |                              |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                   |                              |
| Subject-specific content from the two core areas of digital communication and media technologies is presented in a lecture series, in particular on the basis of current research projects. In the group and under the guidance of a tutorial, students draw connections between the two teaching and research areas. The aim is to emphasise the connection between the two areas at the beginning of the degree course and to promote interface skills.                                                                                                                |                          |                   |                              |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |                   |                              |
| The aim of the interdisciplinary module is to clarify the connection between the core areas of digital communication and media technologies right at the beginning of the degree programme, to convey the added value of interdisciplinary thinking and to promote interface skills. At the same time, students are given an initial understanding of the different disciplinary perspectives on the subjects of the degree programme. The interdisciplinary module thus also forms a basis for the students' own specialisation in the further course of their studies. |                          |                   |                              |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                          |                   |                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |                   |                              |

↑

|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Interdisciplinary Module               |            | Seminar          | german          |

|                                        |                |
|----------------------------------------|----------------|
| <b>Core Area Digital Communication</b> | <b>27 ECTS</b> |
|----------------------------------------|----------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Digital Communication Research                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                 |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1818030                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Module version    |                                 |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | SW-MEW2-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Language          | german                          |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | only in the winter term                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Teaching unit     | Carl-Friedrich-Gauß-Fakultät    |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2 Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Institution       | Department Sozialwissenschaften |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 / 9,0                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Module owner      | Prof. Dr. Monika Tad-dicken     |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 270                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |                                 |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Self studying (h) | 214                             |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1 examination (in the interdisciplinary (MA OGB) or interdisciplinary area (MA Sowi) as course achievementwork): <ul style="list-style-type: none"><li>• 1 term paper (approx. 15 pages) or</li><li>• 1 portfolio (consisting of up to 5 short, complementary parts or comparable achievements such as a poster, debate contribution, exposé, etc. and a written paper (the specific structure can always be found in the seminar plan at the beginning of a course)).</li></ul> |                   |                                 |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |
| In the course "Digital Communication Research" (lecture or seminar), the various social dimensions of digital communication are presented. Forms of digital communication affect practices of everyday information behaviour, cultural participation and relationship management as well as the work of professional communicators in journalism, politics or PR, both within their respective organisations and in relation to their respective communicative reference groups. They change the structures of the social public sphere and break up the dominance of the "mass communication" mode and its institutionalised forms by providing other modes, arenas and levels that make communication content visible and accessible to a (potentially) large group of people. This in turn also has an impact on the understanding of privacy and informational self-determination, because it becomes more difficult to control and delimit audiences for personal information. The lecture provides an overview of the current state of research and discussion. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |
| After completing the module, students will have an overview of the "state of the art" of the utilisation practices, social embedding and effects of digital communication. In addition to social media, this also includes new forms of human-computer interaction and human-machine communication. Students acquire in-depth knowledge of digital communication in selected areas of society and are able to describe and reflect on these from a communication science perspective. They will be able to critically categorise and evaluate the opportunities and risks of various forms of digital communication for individuals and society. Students will thus have the skills required for well-founded theoretical reflection and analysis of digital forms of communication, which are necessary in all professional fields of communication (e.g. journalism, public relations, sales, advertising and marketing) now and in the future.                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                 |



|                                                                                                                                                                                      |            |                  |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                                                                                                               |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                                                                                                               |            |                  |                 |
| (1) Lecture or seminar Digital Communication Research 1<br>(2) Seminar Digital Communication Research 2<br><br>The module starts in the winter semester and runs over two semesters. |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                                                                         |            |                  |                 |
|                                                                                                                                                                                      |            |                  |                 |
| <b>Name of the course</b>                                                                                                                                                            | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Digital Communication Research 1                                                                                                                                                     | 2,0        | Seminar          | german          |
| Digital Communication Research 2                                                                                                                                                     | 2,0        | Seminar          | german          |

|                                                                        |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Title                                                                  | Communication Research Methods and Analyses                                                                                                                                                                                                                                                                                          |                   |                              |
| Number                                                                 | 1818050                                                                                                                                                                                                                                                                                                                              | Module version    |                              |
| Shorttext                                                              | SW-MEW2-05                                                                                                                                                                                                                                                                                                                           | Language          |                              |
| Frequency of offer                                                     | only in the summer term                                                                                                                                                                                                                                                                                                              | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                        | 1 Semester                                                                                                                                                                                                                                                                                                                           | Institution       |                              |
| Hours per Week / ECTS                                                  | 4 / 9,0                                                                                                                                                                                                                                                                                                                              | Module owner      |                              |
| Workload (h)                                                           |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
| Class attendance (h)                                                   | 56                                                                                                                                                                                                                                                                                                                                   | Self studying (h) | 214                          |
| Compulsory requirements                                                |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
| Recommended requirements                                               | Basic knowledge in..<br>• statistics • quantitative and qualitative methods of empirical social research<br>• theories and models of individual and mass communication are assumed                                                                                                                                                   |                   |                              |
| Expected performance/ Type of examination                              | 1 examination:<br><br>• 1 term paper (approx. 15 pages) or<br>• 1 portfolio (consisting of up to 5 short, complementary parts or comparable achievements such as a poster, debate contribution, exposé, etc. and a written paper (the specific structure is always to be taken from the seminar plan at the beginning of a course)). |                   |                              |
| Course achievement                                                     |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
| Contents                                                               |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
|                                                                        |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
| Objective qualification                                                |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
|                                                                        |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
| Literature                                                             |                                                                                                                                                                                                                                                                                                                                      |                   |                              |
| Will be announced in the courses and partly made available in Stud.IP. |                                                                                                                                                                                                                                                                                                                                      |                   |                              |



|                                                                                                                                |            |                  |                 |
|--------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                                                         |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                                                         |            |                  |                 |
| Project seminar (2 SWS) + exercise (1 SWS) + exercise (1 SWS)<br>The courses of the module must be taken in the same semester. |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                   |            |                  |                 |
|                                                                                                                                |            |                  |                 |
| <b>Name of the course</b>                                                                                                      | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Communication Research Methods and Analyses                                                                                    | 2,0        | Seminar          | german          |
| Communication Research Methods and Analyses                                                                                    | 2,0        | Exercise         | german          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Science Communication Research                                                                                                                                                                                                                                                                                                                                                                                                                                       |                   |                                                                            |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1814990                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Module version    |                                                                            |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SW-MEW-99                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Language          | german                                                                     |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | only in the winter term                                                                                                                                                                                                                                                                                                                                                                                                                                              | Teaching unit     |                                                                            |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Institution       | Department Sozialwissenschaften<br>Institut für Kommunikationswissenschaft |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 / 9,0                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Module owner      | Prof. Dr. Monika Tad-dicken                                                |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 270                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                                                                            |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Self studying (h) | 214                                                                        |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                            |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 examination (in the interdisciplinary (MA OGB) or interdisciplinary area (MA Sowi) as coursework): <ul style="list-style-type: none"><li>• 1 term paper (approx. 15 pages) or</li><li>• 1 portfolio (consisting of up to 5 short, complementary parts or comparable achievements such as a poster, debate contribution, exposé, etc. and a written paper (the specific structure can always be found in the seminar plan at the beginning of a course)).</li></ul> |                   |                                                                            |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                            |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                            |
| <p>The course "Fundamentals of Science Communication" (lecture or seminar) provides an overview of the current state of theory and research on the various forms of science communication: journalistic communication about science, interpersonal communication between laypeople about science and the communication of science. The latter can take place externally, i.e. publicly and across fields, or internally, formally or informally.</p> <p>Current research findings are linked with communication science theories. The focus is on theories of content and journalism research, usage, reception and impact research as well as the increasing medialisation of science. This provides a comprehensive insight into the field of research and focuses on the changing understanding of science and the public.</p> <p>In the Science Communication seminar, the relationship between science and the public is examined in depth, focussing in particular on media reporting and lay communication on scientific topics. Current scientific topics alternate as the subject of theoretical and empirical analyses. The aim of the seminar is to analyse both the presentation of a specific scientific topic in the mass media and social communication on and off-line on this topic. In addition to the current state of research, the theories of communication science content and journalism research as well as utilisation, reception and impact research will be used and applied to a specific science communication context.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                            |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                            |
| <p>After completing the module, students will be familiar with the current state of theory and research on science communication. Various forms of science communication are considered, i.e. any external (public), internal or interpersonal communication on scientific topics. Students are thus able to analyse science communication in a theoretically sound manner, they are familiar with current findings from national and international research and can critically categorise communication on scientific topics.</p> <p>These skills are relevant for their own research work and also enable them to carry out tasks focussing on the analytical examination of various forms of science communication.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                            |



| Literature |
|------------|
|            |



| Related courses                                                |     |           |          |
|----------------------------------------------------------------|-----|-----------|----------|
| Rules for the choice of courses                                |     |           |          |
| (1) Lecture or seminar "Fundamentals of Science Communication" |     |           |          |
| (2) Seminar on "Science Communication"                         |     |           |          |
| Compulsory attendance                                          |     |           |          |
|                                                                |     |           |          |
| Name of the course                                             | SWS | Eventtype | Language |
| Basics of Science Communication                                | 2,0 | Lecture   | german   |
| Literature                                                     |     |           |          |
| Literature will be announced at the beginning of the course.   |     |           |          |
| Science Communication (Seminar)                                | 2,0 | Seminar   | german   |

|                                     |                |
|-------------------------------------|----------------|
| <b>Core Area Media Technologies</b> | <b>25 ECTS</b> |
|-------------------------------------|----------------|

|                                                                                                                                                                                                                                                                                                                                                                                      |                         |                   |                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                | Communication Networks  |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                               | 2416660                 | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                            | ET-IDA-66               | Language          | german                                                   |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                   | only in the winter term | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                      | 1                       | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                | 3 / 5,0                 | Module owner      | Prof. Dr. Admela Jukan                                   |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                         | 150                     |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                 | 42                      | Self studying (h) | 108                                                      |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                              |                         |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                            |                         |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                   |                         |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                             |                         |                   |                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                      |                         |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                              |                         |                   |                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                      |                         |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                           |                         |                   |                                                          |
| # Skript # J. F. Kuruse und K. W. Ross, Computer Networking: A Top-Down Approach Featuring the Internet, Addison Wesley, 2005, ISBN: 0-321-26976-4 # W. Stallings, Data and Computer Communications, Pearson Prentise Hall, 2004, ISBN: 0-13-183311-1 # L. L. Peterson und B. S. Davie, Computer Networks: A Systems Approach, Morgan Kaufmann Publishers, 2003, ISBN: 1-55860-833-8 |                         |                   |                                                          |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |

|                                                                                                     |     |          |        |
|-----------------------------------------------------------------------------------------------------|-----|----------|--------|
|                                                                                                     | 2,0 | Lecture  | german |
| <b>Literature</b>                                                                                   |     |          |        |
| * Skript * J.F. Kuruse und K.W. Ross, Computernetze *W. Stallings, Data and Computer Communications |     |          |        |
|                                                                                                     | 1,0 | Exercise | german |
| <b>Literature</b>                                                                                   |     |          |        |
| * Skript * J.F. Kuruse und K.W. Ross, Computernetze *W. Stallings, Data and Computer Communications |     |          |        |

|                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                    |                   |                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                | Human Machine Interaction                                                                                                                                                                                          |                   |                                                 |
| Number                                                                                                                                                                                                                                                                                                                                                                                               | 4225090                                                                                                                                                                                                            | Module version    | V2                                              |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                            | INF-IBR-09                                                                                                                                                                                                         | Language          | german                                          |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                   | only in the summer term                                                                                                                                                                                            | Teaching unit     | Carl-Friedrich-Gauß-Fakultät                    |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                  | Institution       | Institut für Betriebssysteme und Rechnerverbund |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                | 3 / 5,0                                                                                                                                                                                                            | Module owner      | Prof. Dr. Lars Wolf                             |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                         | 150                                                                                                                                                                                                                |                   |                                                 |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                 | 42                                                                                                                                                                                                                 | Self studying (h) | 108                                             |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                    |                   |                                                 |
| Expected performance/<br>Type of examination                                                                                                                                                                                                                                                                                                                                                         | 1 Examination : <ul style="list-style-type: none"><li>• 1 written Exam (90 min) or</li><li>• 1 oral Exam (30 min) or</li><li>• 1 Project Presentation and Documentation or</li><li>• 1 Take-at-Home Exam</li></ul> |                   |                                                 |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                    |                   |                                                 |
| Contents                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                    |                   |                                                 |
| <ul style="list-style-type: none"><li>• Human information processing</li><li>• Design principles and design methods</li><li>• Input and output units for computers</li><li>• Embedded systems and mobile devices</li><li>• Design of user interfaces</li><li>• Design of user interfaces</li><li>• Modelling of user interfaces</li><li>• Evaluation of human-computer interaction systems</li></ul> |                                                                                                                                                                                                                    |                   |                                                 |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                    |                   |                                                 |
| After completing the module, students will have a basic understanding of human-computer interaction. They will have mastered basic techniques for evaluating user interfaces, be familiar with basic rules and techniques for designing user interfaces, and have knowledge of existing user interfaces and their functions.                                                                         |                                                                                                                                                                                                                    |                   |                                                 |
| Literature                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                    |                   |                                                 |
| - Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, Human Computer Interaction                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                    |                   |                                                 |
| Further literature will be announced in the lecture.                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                    |                   |                                                 |



| <b>Related courses</b>                                                             |     |                  |          |
|------------------------------------------------------------------------------------|-----|------------------|----------|
| <b>Rules for the choice of courses</b>                                             |     |                  |          |
|                                                                                    |     |                  |          |
| <b>Compulsory attendance</b>                                                       |     |                  |          |
|                                                                                    |     |                  |          |
| Name of the course                                                                 | SWS | Eventtype        | Language |
|                                                                                    | 3,0 | Lecture/Exercise | german   |
| <b>Literature</b>                                                                  |     |                  |          |
| - Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, Human Computer Interaction |     |                  |          |
| weitere Literatur wird in der Vorlesung bekannt gegeben                            |     |                  |          |

|                                                                                                                                                                                                                                                                                             |                                                                                       |                   |                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|-------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                       | Mobile Communications                                                                 |                   |                                                 |
| Number                                                                                                                                                                                                                                                                                      | 4213400                                                                               | Module version    | V2                                              |
| Shorttext                                                                                                                                                                                                                                                                                   | INF-KM-40                                                                             | Language          | german                                          |
| Frequency of offer                                                                                                                                                                                                                                                                          | only in the summer term                                                               | Teaching unit     | Carl-Friedrich-Gauß-Fakultät                    |
| Module duration                                                                                                                                                                                                                                                                             | 1                                                                                     | Institution       | Institut für Betriebssysteme und Rechnerverbund |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                       | 0 / 5,0                                                                               | Module owner      | Prof. Dr. Lars Wolf                             |
| Workload (h)                                                                                                                                                                                                                                                                                | 150                                                                                   |                   |                                                 |
| Class attendance (h)                                                                                                                                                                                                                                                                        | 56                                                                                    | Self studying (h) | 94                                              |
| Compulsory requirements                                                                                                                                                                                                                                                                     |                                                                                       |                   |                                                 |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                   | 1 graded work: Written exam (90 minutes) or oral exam (20 minutes) or Take-Home-Exam. |                   |                                                 |
| Course achievement                                                                                                                                                                                                                                                                          |                                                                                       |                   |                                                 |
| Contents                                                                                                                                                                                                                                                                                    |                                                                                       |                   |                                                 |
| <ul style="list-style-type: none"><li>- Technical fundamentals of mobile communications</li><li>- Media access</li><li>- Wireless Telecommunications Systems</li><li>- Wireless LANs</li><li>- Network layer aspects</li><li>- Transport Layer Aspects</li><li>- Mobility support</li></ul> |                                                                                       |                   |                                                 |
| Objective qualification                                                                                                                                                                                                                                                                     |                                                                                       |                   |                                                 |
| On completion of the module, students will be familiar with the fundamental challenges and solutions in mobile communications.                                                                                                                                                              |                                                                                       |                   |                                                 |
| Literature                                                                                                                                                                                                                                                                                  |                                                                                       |                   |                                                 |
| <ul style="list-style-type: none"><li>- Jochen Schiller: Mobilkommunikation, Pearson Studium. 2003</li></ul>                                                                                                                                                                                |                                                                                       |                   |                                                 |
| Siehe auch Aktualisierung auf der Webseite der Lehrveranstaltung                                                                                                                                                                                                                            |                                                                                       |                   |                                                 |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |

|                                                                                                               |     |                  |                |
|---------------------------------------------------------------------------------------------------------------|-----|------------------|----------------|
| Mobile Communications                                                                                         | 4,0 | Lecture/Exercise | english german |
| <b>Literature</b>                                                                                             |     |                  |                |
| J. Schiller: Mobilkommunikation - Techniken für das allgegenwärtige Internet, 2. Auflage, Addison-Wesley 2003 |     |                  |                |
|                                                                                                               | 1,0 | Exercise         | english german |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                       |                   |                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Software Engineering 1                                                                |                   |                                                     |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4220430                                                                               | Module version    | V3                                                  |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                      | INF-SSE-43                                                                            | Language          |                                                     |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                             | only in the winter term                                                               | Teaching unit     | Carl-Friedrich-Gauß-Fakultät                        |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1                                                                                     | Institution       | Institut für Softwaretechnik und Fahrzeuginformatik |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 / 5,0                                                                               | Module owner      | Prof. Dr. Wolf-Tilo Balke                           |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 150                                                                                   |                   |                                                     |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                           | 42                                                                                    | Self studying (h) | 108                                                 |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                       |                   |                                                     |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                      | 1 graded work: Written exam (90 minutes) or oral exam (30 minutes) or Take-Home-Exam. |                   |                                                     |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1 Coursework: 50% of homework assignments must be passed.                             |                   |                                                     |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                       |                   |                                                     |
| <ul style="list-style-type: none"><li>- Overview of software techniques</li><li>- Procedures</li><li>- Design, implementation</li><li>- Object orientation</li><li>- Modelling, UML</li><li>- Software/system architectures</li><li>- Patterns in software development</li></ul>                                                                                                                                                                               |                                                                                       |                   |                                                     |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                       |                   |                                                     |
| After completing this module, students will have a fundamental understanding of the development of complex software systems. They will be able to comprehend and model the task at hand and translate it into a design.                                                                                                                                                                                                                                        |                                                                                       |                   |                                                     |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                       |                   |                                                     |
| <ul style="list-style-type: none"><li>- Ian Sommerville: Software Engineering. 7. Aufl. Addison-Wesley, München 2004, ISBN 0-321-21026-3.</li><li>- Helmut Balzert: Lehrbuch der Software-Technik, Spektrum Akademischer Verlag, Heidelberg 1996, 1998, 2001, ISBN 3-8274-0480-0.</li><li>- J. Ludewig, H. Lichter: Software Engineering - Grundlagen, Menschen, Prozesse, Techniken. 1. Auflage. dpunkt-Verlag, Heidelberg 2006, ISBN 3-89864-268-2</li></ul> |                                                                                       |                   |                                                     |





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|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                        | 2,0        | Lecture          | german          |
|                                        | 1,0        | Exercise         | german          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   |                   |                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Probability Theory and Statistics |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2424470                           | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ET-NT-47                          | Language          | german                                                   |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                              | only in the winter term           | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                 | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 / 5,0                           | Module owner      | Prof. Dr. Thomas Kürner                                  |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 150                               |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 56                                | Self studying (h) | 94                                                       |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                   |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                   |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                   |                   |                                                          |
| <ul style="list-style-type: none"><li>• Introduction</li><li>• Fundamentals of probability theory</li><li>• Random variables</li><li>• Special probability distributions</li><li>• Functions of random variables</li><li>• Random processes</li><li>• Transformation of random processes by systems</li></ul>                                                                                                                                                                   |                                   |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |                   |                                                          |
| The lecture provides an understanding of the basic methods of statistics and probability theory. After completing the module, students will have knowledge of the mathematical models used to describe random phenomena. They will be able to solve basic problems in the field of statistics independently.                                                                                                                                                                    |                                   |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   |                   |                                                          |
| ript #<br>A. Papoulis: Probability, random variables, and stochastic processes, McGraw Hill, 1984 #<br>E. Hänsler: Statistische Signale, Springer-Verlag, 2001 #<br>S. Lipschutz: Wahrscheinlichkeitsrechnung - Theorie und Anwendung, McGraw Hill, 1976 #<br>M. Fisz: Wahrscheinlichkeitsrechnung und mathematische Statistik, VEB Deutscher Verlag der Wissenschaften, 1989 #<br>F. Jondral, A. Wiesler, Wahrscheinlichkeitsrechnung und stochastische Prozesse, Teubner 2002 |                                   |                   |                                                          |

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|                                                                                                                                                                                                                                                                                                                                                                                   |            |                  |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                                                                                                                                                                                                                                                                                                            |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                                                                                                                                                                                                                                                                                                            |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                   |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                                                                                                                                                                                                                                                                      |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                   |            |                  |                 |
| <b>Name of the course</b>                                                                                                                                                                                                                                                                                                                                                         | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                                                                                                                                                                                                                                                                                                                                                                   | 2,0        | Exercise         | german          |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                 |            |                  |                 |
| siehe Vorlesung                                                                                                                                                                                                                                                                                                                                                                   |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                   | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                 |            |                  |                 |
| Skript<br>A.Papoulis: Probability, random variables, and stochastic processes, McGraw Hill, 1984<br>E.Hänsler: Statistische Signale, Springer Verlag, 2001<br>S.Lipschutz: Wahrscheinlichkeitsrechnung und mathematische Statistik, VEB Deutscher Verlag der Wissenschaften, 1989<br>F. Jondral, A. Wiesler: Wahrscheinlichkeitsrechnung und stochastische Prozesse, Teubner 2002 |            |                  |                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                         |                   |                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Specialisation Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                         |                   | 23 ECTS                      |
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cloud Computing                                                                                                                                                                                                         |                   |                              |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4223450                                                                                                                                                                                                                 | Module version    |                              |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | INF-VS-45                                                                                                                                                                                                               | Language          | english                      |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | only in the summer term                                                                                                                                                                                                 | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1 Semester                                                                                                                                                                                                              | Institution       |                              |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 / 5,0                                                                                                                                                                                                                 | Module owner      | Rüdiger Kapitza              |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                         |                   |                              |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 56                                                                                                                                                                                                                      | Self studying (h) | 94                           |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                         |                   |                              |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Graded work: written xam, 90 minutes or oral exam, 30 minutes                                                                                                                                                           |                   |                              |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | non-graded work: Successful completion of the homework assignments: Every assignment must be completed with at least 30% of the attainable points, and 50% of the total points across all assignments must be achieved. |                   |                              |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                         |                   |                              |
| <ul style="list-style-type: none"><li>• Overview of Cloud Computing</li><li>• Development of cluster, grid and, utility computing towards Cloud Computing</li><li>• Effects on the economy (cost pressure and energy consumption) and society (e.g. data protection)</li><li>• Fundamentals of distributed programming (Web services/SOAP/REST)</li><li>• Fundamental technology and architecture</li><li>• Virtualization as the basis of Cloud Computing</li><li>• Concepts for hardware virtualization (e.g. Xen, KVM or, VMWare ESX)</li><li>• Advantages and disadvantages of virtualization (e.g. in regards to performance and maintainability)</li><li>• Infrastructure as a Service with the example of Eucalyptus and Amazon EC2</li><li>• Deployment and administration of distributed applications</li><li>• Distributed file systems for cloud applications</li><li>• Provisioning of reliable mass storage based on unreliable components</li><li>• Distributed programming für data-heavy cloud applications</li><li>• Scalable processing of big data</li><li>• Interoperability and multi-cloud</li><li>• Fault-tolerance and security in a cloud computing context</li><li>• Current research trends (e.g. 'new' programming languages, intrusion-resistant systems)</li></ul> |                                                                                                                                                                                                                         |                   |                              |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                         |                   |                              |
| After completing this module, the students know the fundamentals, methods and, techniques of Cloud Computing. Further, the students know existing Cloud Computing techniques and can develop and assess applications in this setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                         |                   |                              |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                         |                   |                              |
| <ul style="list-style-type: none"><li>• A view of cloud computing M. Armbrust, A. Fox, R. Griffith, A. D. Joseph, R. Katz, A. Konwinski, G. Lee, D. Patterson, A. Rabkin, I. Stoica, and M. Zaharia. A view of cloud computing. Communication of the ACM, 53(4):50-58, 2010. Cloud computing: An overview M. Creeger.</li><li>• Cloud computing: An overview.Oueue, 7(5):3-4, 2009. Advisor-Creeger, Mache.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                         |                   |                              |

Weitere Literaturangaben siehe unter <http://www.ibr.cs.tu-bs.de/courses/>



| <b>Related courses</b>                 |     |                              |                |
|----------------------------------------|-----|------------------------------|----------------|
| <b>Rules for the choice of courses</b> |     |                              |                |
|                                        |     |                              |                |
| <b>Compulsory attendance</b>           |     |                              |                |
|                                        |     |                              |                |
| Name of the course                     | SWS | Eventtype                    | Language       |
|                                        | 2,0 | Lecture                      | english german |
|                                        | 2,0 | Exercise                     | english german |
|                                        | 1,0 | Online-exercise, small group | german         |

|                                                                                                                                                                                                                                                 |                                              |                   |                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                           | Coding Theory                                |                   |                                                          |
| Number                                                                                                                                                                                                                                          | 2424420                                      | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                       | ET-NT-42                                     | Language          | english german                                           |
| Frequency of offer                                                                                                                                                                                                                              | only in the summer term                      | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                 | 1                                            | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                           | 4 / 5,0                                      | Module owner      | Prof. Dr. Thomas Kürner                                  |
| Workload (h)                                                                                                                                                                                                                                    | 150                                          |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                            | 56                                           | Self studying (h) | 94                                                       |
| Compulsory requirements                                                                                                                                                                                                                         |                                              |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                       | Oral exam (20 min) or written exam (120 min) |                   |                                                          |
| Course achievement                                                                                                                                                                                                                              | Colloquium or lab journal                    |                   |                                                          |
| Contents                                                                                                                                                                                                                                        |                                              |                   |                                                          |
|                                                                                                                                                                                                                                                 |                                              |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                         |                                              |                   |                                                          |
|                                                                                                                                                                                                                                                 |                                              |                   |                                                          |
| Literature                                                                                                                                                                                                                                      |                                              |                   |                                                          |
| Lecture notes<br>H.Rohling: Einführung in die Informations- und Codierungstheorie, Teubner<br>R.Togneri, C.J.S. deSilva: Fundamentals of Information Theory and Coding Design, Chapman&Hall/CRC<br>H.Schneider-Obermann: Kanalcodierung, Vieweg |                                              |                   |                                                          |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |

|                                                                                                                                                                                                                                                          |     |            |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------|----------------|
| Coding Theory                                                                                                                                                                                                                                            | 2,0 | Lecture    | english german |
| <b>Literature</b>                                                                                                                                                                                                                                        |     |            |                |
| Vorlesungsskript<br>H. Rohling: Einführung in die Informations- und Codierungstheorie, Teubner<br>R. Togneri, C. J. S. de Silva: Fundamentals of Information Theory and Coding Design, Chapman&Hall/CRC<br>H. Schneider-Obermann: Kanalcodierung, Vieweg |     |            |                |
| Coding Theory                                                                                                                                                                                                                                            | 1,0 | Exercise   | english german |
| <b>Literature</b>                                                                                                                                                                                                                                        |     |            |                |
| siehe Vorlesung                                                                                                                                                                                                                                          |     |            |                |
| Computer exercise on Coding Theory                                                                                                                                                                                                                       | 1,0 | Laboratory | english german |

|                                                                                                                     |                                                                                       |                   |                              |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|------------------------------|
| Title                                                                                                               | Computer Graphics - Introduction                                                      |                   |                              |
| Number                                                                                                              | 4216300                                                                               | Module version    | V2                           |
| Shorttext                                                                                                           | INF-CG-30                                                                             | Language          |                              |
| Frequency of offer                                                                                                  | only in the winter term                                                               | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                                                                     | 1                                                                                     | Institution       |                              |
| Hours per Week / ECTS                                                                                               | 4 / 5,0                                                                               | Module owner      | Prof. Dr. Marcus Magnor      |
| Workload (h)                                                                                                        | 150                                                                                   |                   |                              |
| Class attendance (h)                                                                                                | 56                                                                                    | Self studying (h) | 94                           |
| Compulsory requirements                                                                                             |                                                                                       |                   |                              |
| Expected performance/ Type of examination                                                                           | 1 graded work: Written exam (90 minutes) or oral exam (30 minutes) or Take-Home-Exam. |                   |                              |
| Course achievement                                                                                                  |                                                                                       |                   |                              |
| Contents                                                                                                            |                                                                                       |                   |                              |
|                                                                                                                     |                                                                                       |                   |                              |
| Objective qualification                                                                                             |                                                                                       |                   |                              |
|                                                                                                                     |                                                                                       |                   |                              |
| Literature                                                                                                          |                                                                                       |                   |                              |
| - James Foley, AndriesVan Dam, et al., Computer Graphics: Principles and Practice, 2. Ausgabe, Addison-Wesley, 2009 |                                                                                       |                   |                              |
| - Peter Shirley: Realistic Ray-Tracing. AK Peters, 2009                                                             |                                                                                       |                   |                              |
| - Peter Shirley, Steve Marschner: Fundamentals of Computer Graphics. AK Peters/CRC Press, 2009.                     |                                                                                       |                   |                              |

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|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |                  |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------|--------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4,0 | Lecture/Exercise | german |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                  |        |
| <ul style="list-style-type: none"><li>- Peter Shirley: Realistic Ray-Tracing . AK Peters, 2009</li><li>- James Foley, Andries Van Dam, et. Al., Computer Graphics: Principles and Practice. Addison-Wesley, 2009</li><li>- Frank Nielsen, Visual Computing, Charles River Media, 2005</li><li>- Steven J. Gortler, Foundations of 3D Graphics, Mit Pr, 2012</li><li>- John F. Hughes, Computer Graphics: Principles and Practice, Addison-Wesley, 2009</li></ul> |     |                  |        |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                         | Computer Networks 2                                                                   |                   |                                                           |
| Number                                                                                                                                                                                                                                                                                        | 4213390                                                                               | Module version    | V2                                                        |
| Shorttext                                                                                                                                                                                                                                                                                     | INF-KM-39                                                                             | Language          | english german                                            |
| Frequency of offer                                                                                                                                                                                                                                                                            | only in the winter term                                                               | Teaching unit     | Carl-Friedrich-Gauß-Fakultät                              |
| Module duration                                                                                                                                                                                                                                                                               | 1                                                                                     | Institution       | Institut für Betriebssy-<br>steme und Rechnerver-<br>bund |
| Hours per Week /<br>ECTS                                                                                                                                                                                                                                                                      | 0 / 5,0                                                                               | Module owner      | Prof. Dr. Lars Wolf                                       |
| Workload (h)                                                                                                                                                                                                                                                                                  | 150                                                                                   |                   |                                                           |
| Class attendance<br>(h)                                                                                                                                                                                                                                                                       | 56                                                                                    | Self studying (h) | 94                                                        |
| Compulsory<br>requirements                                                                                                                                                                                                                                                                    |                                                                                       |                   |                                                           |
| Recommended<br>requirements                                                                                                                                                                                                                                                                   | INF 2230 (Computer Networks) or equivalent knowledge                                  |                   |                                                           |
| Expected<br>performance/<br>Type of examina-<br>tion                                                                                                                                                                                                                                          | 1 graded work: Written exam (90 minutes) or oral exam (20 minutes) or Take-Home-Exam. |                   |                                                           |
| Course<br>achievement                                                                                                                                                                                                                                                                         |                                                                                       |                   |                                                           |
| Contents                                                                                                                                                                                                                                                                                      |                                                                                       |                   |                                                           |
| - Internet protocols<br>- IP<br>- TCP<br>- Routing methods<br>- Newer protocols and methods                                                                                                                                                                                                   |                                                                                       |                   |                                                           |
| Objective qualification                                                                                                                                                                                                                                                                       |                                                                                       |                   |                                                           |
| On completion of this module, students will have deepened their knowledge from the course 'Computer Networks 1'. They will be familiar with the processes used on the Internet and the procedures involved.                                                                                   |                                                                                       |                   |                                                           |
| Literature                                                                                                                                                                                                                                                                                    |                                                                                       |                   |                                                           |
| Andrew Tanenbaum, David Wetherall, Nick Feamster, Computer Networks, 6.Ed. 2021, Print-ISBN: 978-1-292-37406-2, E-ISBN: 978-1-292-37401-7<br>James Kurose, Keith Ross. Computer Networking. A Top-Down Approach, 2021, 8th edition, Print-ISBN: 978-1-292-40546-9, E-ISBN: 978-1-292-40551-3. |                                                                                       |                   |                                                           |



| <b>Related courses</b>                                                                                                                                                                                                                                                                                                                    |     |                  |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------|----------|
| <b>Rules for the choice of courses</b>                                                                                                                                                                                                                                                                                                    |     |                  |          |
|                                                                                                                                                                                                                                                                                                                                           |     |                  |          |
| <b>Compulsory attendance</b>                                                                                                                                                                                                                                                                                                              |     |                  |          |
|                                                                                                                                                                                                                                                                                                                                           |     |                  |          |
| Name of the course                                                                                                                                                                                                                                                                                                                        | SWS | Eventtype        | Language |
|                                                                                                                                                                                                                                                                                                                                           | 4,0 | Lecture/Exercise | english  |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                         |     |                  |          |
| - Andrew S. Tanenbaum; David J. Wetherall: Computer Networks. International Edition. 5th edition. Pearson, 2010. ISBN-10: 0132553171 / ISBN-13: 9780132553179 - James F. Kurose; Keith W. Ross: Computer Networking: A Top-Down Approach. International Edition. 6th edition. Pearson, 2012. ISBN-10: 0273768964 / ISBN-13: 9780273768968 |     |                  |          |
|                                                                                                                                                                                                                                                                                                                                           | 2,0 | Exercise         | english  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                           |                   |                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Digital Signal Transmission and Computational Experiments |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2424670                                                   | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ET-NT-67                                                  | Language          | german                                                   |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | only in the summer term                                   | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                                                         | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 8 / 10,0                                                  | Module owner      | Prof. Dr. Eduard Jorswieck                               |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 300                                                       |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 112                                                       | Self studying (h) | 188                                                      |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                           |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Oral exam (30 min) or written exam (180 min)              |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Colloquium or lab minutes                                 |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                           |                   |                                                          |
| <p>Part I:</p> <ul style="list-style-type: none"><li>- Deterministic signals in LTI systems</li><li>- Fourier transform</li><li>- Discrete signals and systems</li><li>- Correlation function of deterministic signals</li><li>- System theory of low-pass and band-pass systems</li></ul> <p>Part II:</p> <ul style="list-style-type: none"><li>- Statistical signal processing</li><li>- Multiplexing</li><li>- Binary transmission with low-pass signals</li><li>- Binary transmission with band-pass signals</li><li>- Digital modulation</li></ul> |                                                           |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                           |                   |                                                          |
| The students describe signal processing systems by their impulse response or transfer function. They know the fundamental principles of digital communications systems.<br>Practical knowledge and examples in computational experiments.                                                                                                                                                                                                                                                                                                               |                                                           |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                           |                   |                                                          |
| <ul style="list-style-type: none"><li>- Ohm, Lüke: Signalübertragung, Springer-Verlag, ISBN 3-540-67768-2</li><li>- U.Reimers: Digitale Fernsehtechnik, 2.Aufl. 1997, ISBN 3-540-60945-8</li></ul>                                                                                                                                                                                                                                                                                                                                                      |                                                           |                   |                                                          |
| Remark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                           |                   |                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                           |                   |                                                          |



|                                                                                                                                  |            |                  |                 |
|----------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                                                           |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                                                           |            |                  |                 |
|                                                                                                                                  |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                     |            |                  |                 |
|                                                                                                                                  |            |                  |                 |
| <b>Name of the course</b>                                                                                                        | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                                                                                                                  | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                                                                |            |                  |                 |
| Ohm, Lüke: Signalübertragung, Springer-Verlag, ISBN 3-540-67768-2 Reimers: Digitale Fernsehtechnik, 2. Aufl., ISBN 3-540-60945-8 |            |                  |                 |
|                                                                                                                                  | 2,0        | Laboratory       | german          |
| <b>Literature</b>                                                                                                                |            |                  |                 |
| siehe Vorlesung                                                                                                                  |            |                  |                 |
|                                                                                                                                  | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                                                                |            |                  |                 |
| Ohm, Lüke: Signalübertragung, Springer-Verlag, ISBN 3-540-67768-2 Reimers: Digitale Fernsehtechnik, 2. Aufl., ISBN 3-540-60945-8 |            |                  |                 |
|                                                                                                                                  | 1,0        | Exercise         | german          |
| <b>Literature</b>                                                                                                                |            |                  |                 |
| siehe Vorlesung                                                                                                                  |            |                  |                 |
|                                                                                                                                  | 1,0        | Exercise         | german          |
| <b>Literature</b>                                                                                                                |            |                  |                 |
| siehe Vorlesung                                                                                                                  |            |                  |                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                       |                   |                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|-----------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Distributed data management                                                           |                   |                                   |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4214580                                                                               | Module version    | V2                                |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | INF-IS-58                                                                             | Language          | german                            |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | only in the winter term                                                               | Teaching unit     | Carl-Friedrich-Gauß-Fakultät      |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                                                                     | Institution       | Institut für Informations-systeme |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0 / 5,0                                                                               | Module owner      | Prof. Dr. Wolf-Tilo Balke         |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 150                                                                                   |                   |                                   |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 42                                                                                    | Self studying (h) | 108                               |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                       |                   |                                   |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1 graded work: Written exam (90 minutes) or oral exam (30 minutes) or Take-Home-Exam. |                   |                                   |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 Coursework: 50% of the exercises must be passed                                     |                   |                                   |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                       |                   |                                   |
| <ul style="list-style-type: none"><li>- Distributed database architectures and data distribution</li><li>- Vertical and horizontal fragmentation</li><li>- Distributed query processing</li><li>- Distributed transactions</li><li>- Fundamentals of parallel database systems</li><li>- Parallel query processing</li><li>- Fundamentals of peer-to-peer networks</li><li>- Random graphs, small worlds and scale-free networks</li><li>- Structured networks with distributed hash tables</li><li>- Schema-based peer-to-peer networks</li><li>- Information retrieval in peer-to-peer networks</li></ul> |                                                                                       |                   |                                   |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                       |                   |                                   |
| After completing this module, students will have acquired fundamental knowledge in the areas of distributed database systems and peer-to-peer data management.                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |                   |                                   |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                       |                   |                                   |
| <ul style="list-style-type: none"><li>- Peter Mahlmann, Christian Schindelhauer: P2P Netzwerke. Springer Verlag, 2007.</li><li>- Ralf Steinmetz, Klaus Wehrle: Peer-to-Peer Systems and Applications. Springer Verlag, 2005.</li><li>- M. Tamer Ozsu, Patrick Valduriez: Principles of Distributed Data Systems. Prentice Hall, 1997.</li></ul>                                                                                                                                                                                                                                                             |                                                                                       |                   |                                   |



| <b>Related courses</b>                 |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                        | 2,0        | Lecture/Exercise | german          |
|                                        | 1,0        | Exercise         | german          |

|                                                                                                                                                                                                                                                                                                                                                                                                                |                                                      |                   |                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                          | Fundamentals of Digital Signal Processing            |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                                                         | 2424480                                              | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                      | ET-NT-48                                             | Language          | german                                                   |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                             | every term                                           | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                | 1                                                    | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                          | 3 / 5,0                                              | Module owner      | Prof. Dr. Tim Fingscheidt                                |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                   | 150                                                  |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                           | 42                                                   | Self studying (h) | 108                                                      |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                        |                                                      |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                      | Written exam (120 minutes) or oral exam (30 minutes) |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                             |                                                      |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |                   |                                                          |
| <div>- Discrete-time signals and systems #<br/>- Fourier transforms #<br/>- Z-transforms and applications #<br/>- Discrete-time IIR filter design #<br/>- Discrete-time FIR filter design #<br/>- Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT) #<br/>- Basics of multi-rate processing and filter banks</div>                                                                             |                                                      |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                        |                                                      |                   |                                                          |
| After completing this module, students will have basic knowledge on the tools of digital signal processing in the time and frequency domain and can apply these tools to corresponding problems.                                                                                                                                                                                                               |                                                      |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                     |                                                      |                   |                                                          |
| <div>- Lecture notes<br/>- A.V. Oppenheim, R.W. Schafer, J.R. Buck: "Zeitdiskrete Signalverarbeitung" , Pearson Verlag, 2004<br/>- K.D. Kammeyer, K. Kroschel: "Digitale Signalverarbeitung" , Teubner Verlag, 2002<br/>- A.V. Oppenheim, R.W. Schafer, J.R. Buck: "Discrete Time Signal Processing" , Prentice-Hall, 2004<br/>- H.-W. Schüßler: "Digitale Signalverarbeitung 1" , Springer Verlag, 1994</div> |                                                      |                   |                                                          |
| Remark                                                                                                                                                                                                                                                                                                                                                                                                         |                                                      |                   |                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                |                                                      |                   |                                                          |





| <b>Related courses</b>                                                                                                                                                                                                                                                                                                                                                                                                                     |            |                  |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Rules for the choice of courses</b>                                                                                                                                                                                                                                                                                                                                                                                                     |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                                                                                                                                                                                                                                                                                                                               |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |                 |
| <b>Name of the course</b>                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |                 |
| <ul style="list-style-type: none"> <li>• A. V. Oppenheim, R. W. Schafer, J. R. Buck: Zeitdiskrete Signalverarbeitung, Pearson Studium, 2004</li> <li>• K. D. Kammeyer, K. Kroschel: Digitale Signalverarbeitung, Teubner Verlag, 2002</li> <li>• A. V. Oppenheim, R. W. Schafer, J. R. Buck: Discrete Time Signal Processing, Prentice Hall, 2004</li> <li>• H.-W. Schüßler: Digitale Signalverarbeitung, Springer Verlag, 1994</li> </ul> |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1,0        | Exercise         | german          |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |                 |
| siehe Vorlesung                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |                 |

|                                                                                                                                                                                                                                                                                                                                                                  |                                             |                   |                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                            | Fundamentals of Mobile Radio Communications |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                           | 2424490                                     | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                        | ET-NT-49                                    | Language          | english german                                           |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                               | only in the winter term                     | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                  | 1                                           | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                            | 4 / 5,0                                     | Module owner      | Prof. Dr. Thomas Kürner                                  |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                     | 150                                         |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                             | 56                                          | Self studying (h) | 94                                                       |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                          |                                             |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                        | Oral exam (20 min) or written exam (90 min) |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                               |                                             |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                         |                                             |                   |                                                          |
| 1. Introduction<br>2. Wave Propagation<br>3. Radio Transmission<br>4. Media Access<br>5. 3GPP Wireless Systems<br>6. IEEE 802 Wireless Systems                                                                                                                                                                                                                   |                                             |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                          |                                             |                   |                                                          |
| The lecture provides the basics in the areas of the air interface of mobile communication systems. Students will acquire knowledge on the structure and functionality of cellular and wireless local area networks.                                                                                                                                              |                                             |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                       |                                             |                   |                                                          |
| <ul style="list-style-type: none"><li>• Lecture notes</li><li>• C. Lüders, Mobilfunksysteme, Vogel-Verlag 2001</li><li>• # J. Schiller, Mobilkommunikation, Addison-Wesley 2000</li><li>• N. Geng, W. Wiesbeck, Planungsmethoden für die Mobilkommunikation, Springer-Verlag 1998 #</li><li>• A. Molisch, Wireless Communications, Addison-Wesley 2005</li></ul> |                                             |                   |                                                          |
| Remark                                                                                                                                                                                                                                                                                                                                                           |                                             |                   |                                                          |
| This Master's module is also suitable for Bachelor students.                                                                                                                                                                                                                                                                                                     |                                             |                   |                                                          |



| <b>Related courses</b>                                                                                                                                                                                                                                      |     |           |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|----------------|
| <b>Rules for the choice of courses</b>                                                                                                                                                                                                                      |     |           |                |
|                                                                                                                                                                                                                                                             |     |           |                |
| <b>Compulsory attendance</b>                                                                                                                                                                                                                                |     |           |                |
|                                                                                                                                                                                                                                                             |     |           |                |
| Name of the course                                                                                                                                                                                                                                          | SWS | Eventtype | Language       |
|                                                                                                                                                                                                                                                             | 1,5 | Exercise  | english german |
| <b>Literature</b>                                                                                                                                                                                                                                           |     |           |                |
| siehe Vorlesung                                                                                                                                                                                                                                             |     |           |                |
|                                                                                                                                                                                                                                                             | 2,5 | Lecture   | english german |
| <b>Literature</b>                                                                                                                                                                                                                                           |     |           |                |
| Skript C. Lüders, Mobilfunksysteme, Vogel-Verlag 2001 J. Schiller, Mobilkommunikation, Addison-Wesley 2000 N. Geng, W. Wiesbeck, Planungsmethoden für die Mobilkommunikation, Springer-Verlag 1998 A. Molisch, Wireless Communications, Addison-Wesley 2005 |     |           |                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                             |                   |                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Information Theory                          |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2424720                                     | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ET-NT-72                                    | Language          | german                                                   |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | only in the winter term                     | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                           | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3 / 5,0                                     | Module owner      | Prof. Dr. Eduard Jorswieck                               |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 150                                         |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 42                                          | Self studying (h) | 108                                                      |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                             |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Written exam (90 min) or oral exam (30 min) |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                             |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                             |                   |                                                          |
| <ul style="list-style-type: none"><li>Basics from probability theory<ul style="list-style-type: none"><li>Event, probability, random variable, random vector, stochastic process, convergence of random series, convergence theorems</li></ul></li><li>Basics from information theory<ul style="list-style-type: none"><li>Measures for discrete random variables: entropy, conditional entropy, relativ entropy, mutual information, conditional mutual information, inequalities</li><li>Measures for continous random variables: differential entropy, conditional differential entropy, relative entropy, mutual information, inequalities</li><li>Measure for random series</li><li>Typical sequences and asymptotic equipartition property</li></ul></li><li>Source and source coding<ul style="list-style-type: none"><li>Definition and properties</li><li>Source coding for discrete memoryless sources (fixed and variable-length)</li><li>Selected source codes: Morse, Huffman, Shannon-Fano-Elias</li></ul></li><li>Data transmission and channel capacity<ul style="list-style-type: none"><li>Discrete memoryless channel: channel coding theorem</li><li>Discrete memoryless channel with state: channel capacities</li><li>Gaussian channel: model and channel coding theorem</li><li>Bandlimited Gaussian channel, vector valued channels</li></ul></li></ul> |                                             |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                             |                   |                                                          |
| The lecture provides an introduction to the fundamentals of Shannon information theory. The goal is that students can derive the main information theoretic results on maximal achievable lossless (source coding) and lossy (rate distortion theory) compression of data and on maximum data rates for reliable data transmission (channel coding). The methods and tools required, e.g., information measures (entropy, mutual information, capacity etc.) and their properties (typical sequences) will be covered as well as practical applicable simple codes (block, turbo and polar codes).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                             |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                             |                   |                                                          |
| #R.W. Yeung: Information Theory and Network Coding, Part I, Springer, 2008.<br>R.W. Yeung: A First Course in Information Theory, Springer, 2002.<br>T.M. Cover und J.A. Thomas: Elements of Information Theory. Wiley-Interscience. 2006.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                             |                   |                                                          |

R.G. Gallager: Information Theory and Reliable Communication, Wiley, 1968.  
 R.G. Gallager: Principles of Digital Communication, Cambridge University Press, 2008.  
 S. Moser: S. Moser: Information Theory, <https://moser-isi.ethz.ch/scripts.html#it>



| Related courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |           |          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|----------|
| Rules for the choice of courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |           |          |
| Compulsory attendance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |           |          |
| Name of the course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | SWS | Eventtype | Language |
| Information Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2,0 | Lecture   | german   |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |           |          |
| - R.W. Yeung: Information Theory and Network Coding, Part I, Springer, 2008. - R.W. Yeung: A First Course in Information Theory, Springer, 2002. - T.M. Cover und J.A. Thomas: Elements of Information Theory, Wiley-Interscience, 2006. - R.G. Gallager: Information Theory and Reliable Communication, Wiley, 1968. - R.G. Gallager: Principles of Digital Communication, Cambridge University Press, 2008. - S. Moser: S. Moser: Information Theory, <a href="https://moser-isi.ethz.ch/scripts.html#it">https://moser-isi.ethz.ch/scripts.html#it</a> |     |           |          |
| Information Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1,0 | Exercise  | german   |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |           |          |
| - R.W. Yeung: Information Theory and Network Coding, Part I, Springer, 2008. - R.W. Yeung: A First Course in Information Theory, Springer, 2002. - T.M. Cover und J.A. Thomas: Elements of Information Theory, Wiley-Interscience, 2006. - R.G. Gallager: Information Theory and Reliable Communication, Wiley, 1968. - R.G. Gallager: Principles of Digital Communication, Cambridge University Press, 2008. - S. Moser: S. Moser: Information Theory, <a href="https://moser-isi.ethz.ch/scripts.html#it">https://moser-isi.ethz.ch/scripts.html#it</a> |     |           |          |

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| Title                                                                                                                                                                                                                                                                                                                                                                                                | Media Technologies      |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                                               | 1814270                 | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                            | ET-NT-27                | Language          | german                                                   |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                   | only in the summer term | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                      | 2                       | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                | 4 / 6,0                 | Module owner      | Prof. Dr. Ulrich Reimers                                 |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                         | 180                     |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                 | 56                      | Self studying (h) | 124                                                      |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                              |                         |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                            |                         |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                   |                         |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                                             |                         |                   |                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                      |                         |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                              |                         |                   |                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                      |                         |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                                           |                         |                   |                                                          |
| Bildkommunikation II: # U.Reimers: DVB-Digitale Fernsehetechnik: Datenkompression und Übertragung, Springer Verlag, 3. Auflage, 2008 Elektroakustik: # Zoller/Zwicker: Elektroakustik, Springer Verlag # Kuttruff: Akustik - Eine Einführung, S.Hirzel Verlag Stuttgart Leipzig # Cremer/Möser: Technische Akustik, Springer Verlag # Ahnert: Beschallungstechnik, S.Hirzel Verlag Stuttgart Leipzig |                         |                   |                                                          |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |

|                                                                                                                                                                                                                                                                                                                                                                                               |     |         |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|--------|
|                                                                                                                                                                                                                                                                                                                                                                                               | 2,0 | Lecture | german |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                             |     |         |        |
| H.Lang: Farbmeterik und Farbfernsehen, Oldenbourg Verlag, 1978 R.Mäusl: Fernsehetechnik, Hüthig Verlag, 1995 U.Reimers: Digitale Fernsehetechnik - Datenkompression und Übertragungstechnik, Springer Verlag, 3. Auflage, 2007 U.Schmidt: Professionelle Videotechnik, Springer Verlag 2000 A.N.Netravali, B.G.Haskell: Digital Pictures - Representation and Compression, Plenum Press, 1991 |     |         |        |
|                                                                                                                                                                                                                                                                                                                                                                                               | 2,0 | Lecture | german |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                             |     |         |        |
| Zoller/Zwicker: Elektroakustik, Springer Verlag Kuttruff: Akustik - Eine Einführung, S.Hirzel Verlag Stuttgart Leipzig Cremer/Möser: Technische Akustik, Springer Verlag Ahnert: Beschallungstechnik, S.Hirzel Verlag Stuttgart Leipzig                                                                                                                                                       |     |         |        |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Media Analysis                                                                                                                                                                                                                                                                                          |                   |                              |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2199310                                                                                                                                                                                                                                                                                                 | Module version    |                              |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SW-MEW-91                                                                                                                                                                                                                                                                                               | Language          | german                       |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | every term                                                                                                                                                                                                                                                                                              | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1 Semester                                                                                                                                                                                                                                                                                              | Institution       |                              |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 6 / 9,0                                                                                                                                                                                                                                                                                                 | Module owner      | Heike Klippel                |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                         |                   |                              |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 90                                                                                                                                                                                                                                                                                                      | Self studying (h) | 180                          |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                         |                   |                              |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1 examination (graded): <ul style="list-style-type: none"><li>• 1 term paper (12 to 15 pages) and/or</li><li>• 1 presentation with written paper (10 to 12 pages) and/or</li><li>• 1 written examination (1 x 180 minutes or 3 x 60 minutes) and/or</li><li>• 1 oral examination (30 minutes)</li></ul> |                   |                              |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                         |                   |                              |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                         |                   |                              |
| The subject of the module is media analysis and media interpretation using examples from as broadly differentiated areas of media production as possible. The material for formal, content-related and discursive analyses and interpretations should come from different media, genres, historical periods and cultural circles. A broad spectrum of methodological diversity will be used, and the results of analytical investigations will be subjected to reflection and interpretation and placed in broader (inter-)disciplinary contexts. The lecturers' current objects of research and methods of analysis are systematically incorporated into teaching and module examinations, e.g. in the past from the DFG project "The Poison Motif in Film" in the course of the same name. |                                                                                                                                                                                                                                                                                                         |                   |                              |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                         |                   |                              |
| The aim is to achieve a differentiated understanding and increased aesthetic sensitisation to a wide variety of media forms. In addition, the methods and analytical approaches learnt in the BA are deepened and practised for the purpose of transferring their results into plausibly argued interpretation and clear linguistic presentation. The interest in knowledge is focussed on levels of depth and contextual meanings of media products. Discursive forms of learning such as presentations, group work, seminar discussions, predominantly based on media reception. The exercise serves to deepen individual key topics.                                                                                                                                                      |                                                                                                                                                                                                                                                                                                         |                   |                              |
| The students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                         |                   |                              |
| <ul style="list-style-type: none"><li>• know, remember and understand current concepts, procedures and methods for analysing media products and contexts and can apply and evaluate them to media objects and phenomena.</li><li>• know, remember and understand current research projects and are able to relate them to what they have learnt and actively participate in them if necessary.</li><li>• can clearly present, communicate, acquire and critically evaluate knowledge discursively by means of presentation, discussion and argumentation, including cooperative forms of learning and group work.</li></ul>                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                         |                   |                              |
| Professional Competences: Students are able to understand, classify, evaluate and critically reflect on media studies analyses and current research questions and designs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                         |                   |                              |



Interdisciplinary Competences: Students are able to understand and evaluate analytical perspectives and to analyse complex content, objects and constellations alone and in collaborative groups and to present, discuss, appropriate and critically evaluate the results.

### Literature



### Related courses

#### Rules for the choice of courses

[1] 2xSE

[2] 1xUE

#### Compulsory attendance

Work performance for the award of credits and grades:

Active participation in the courses: Lecture (lecturer's presentation; preparation and follow-up, notes, self-study), seminar (literature study, presentation, discussion) and passing the examinations. In order to achieve the qualification objectives for the seminars and exercises, it is necessary for the participants to take part in the interactive discussions and group work within the course. This requires the continuous presence of the participants, i.e. usually at least 80% of the attendance time.

| Name of the course                   | SWS | Eventtype | Language |
|--------------------------------------|-----|-----------|----------|
| Masters Seminar Media Analysis (HBK) | 2,0 | Seminar   | german   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                        |                   |                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Media Theory and Media History                                                                                                                                                                                                                                                                         |                   |                              |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2199320                                                                                                                                                                                                                                                                                                | Module version    |                              |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MW-STD-32                                                                                                                                                                                                                                                                                              | Language          | german                       |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | every term                                                                                                                                                                                                                                                                                             | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 Semester                                                                                                                                                                                                                                                                                             | Institution       |                              |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 6 / 9,0                                                                                                                                                                                                                                                                                                | Module owner      | Ulrike Bergermann            |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                        |                   |                              |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 90                                                                                                                                                                                                                                                                                                     | Self studying (h) | 180                          |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                        |                   |                              |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1 Examination results: <ul style="list-style-type: none"><li>• 1 term paper (12 to 15 pages) and/or</li><li>• 1 presentation with written paper (10 to 12 pages) and/or</li><li>• 1 written examination (1 x 180 minutes or 3 x 60 minutes) and/or</li><li>• 1 oral examination (30 minutes)</li></ul> |                   |                              |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                        |                   |                              |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                        |                   |                              |
| Selected theoretical complexes and discourses are used to reflect on specific problems in media theory. Possible topics include cultural studies/history and theory of mass media, digitality as a 'meta-medium' in media history, media and subjective experience, changes in the concept of the public sphere, transparency vs. manipulation in different phases of media history, comparative approaches to reception theory. The lecturers' current research topics, for example from the fields of game studies, film studies or postcolonial studies, are systematically incorporated into teaching and module examinations, e.g. in the past on the DFG project "Kulturtechnik Unternehmensplanspiel" in the advanced seminar "Kulturtechnik Spiel". The media-theoretical core can be expanded in two directions: on the one hand by adding basic and background texts that are relevant to the respective media-theoretical approaches, and on the other hand by transferring them to meta-levels with a philosophical orientation. Media-historical approaches are analysed and interpreted with regard to their theoretical implications. Theories are given a historical perspective, history is interpreted as relevant to theory. |                                                                                                                                                                                                                                                                                                        |                   |                              |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                        |                   |                              |
| The handling of theoretical and historical texts should now go beyond their purely content-related level and be deepened, above all with regard to the development of competences for the independent and critical discussion of media science issues and research designs. Students should no longer be able to deal with media studies problems within a narrow framework, but to place them in broader contexts and analyse them at a deeper level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                        |                   |                              |
| The students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                        |                   |                              |
| <ul style="list-style-type: none"><li>• know, remember and understand current media theoretical and historical topics, approaches and concepts and are able to analyse, relate and evaluate them.</li><li>• know, remember and understand current research projects and are able to relate them to what they have learnt and, if necessary, actively participate in them.</li><li>• can communicate, acquire and critically evaluate knowledge discursively by means of presentation, discussion and argumentation, including cooperative forms of learning and group work.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                        |                   |                              |

Professional Competences: Students are able to understand, categorise, evaluate and critically reflect on media studies theory formation and historiography as well as current research questions and designs.

Interdisciplinary Competences: Students are able to understand complex theories and historical issues and to present, discuss, appropriate and critically evaluate complex content alone and in collaborative groups.

### Literature



### Related courses

#### Rules for the choice of courses

[1] 2xSE  
[2] 1xUE

#### Compulsory attendance

Work performance for the award of credits and grades:

Active participation in the courses: Lecture (lecturer's presentation; preparation and follow-up, notes, self-study), seminar (literature study, presentation, discussion) and passing the examinations. In order to achieve the qualification objectives for the seminars and exercises, it is necessary for the participants to take part in the interactive discussions and group work within the course. This requires the continuous presence of the participants, i.e. usually at least 80% of the attendance time.

| Name of the course                                  | SWS | Eventtype | Language |
|-----------------------------------------------------|-----|-----------|----------|
| Master Seminar Media Theory and History Event (HBK) | 2,0 | Seminar   | german   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                  |                   |                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Modelling and Simulation of Mobile Radio Systems |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2424400                                          | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ET-NT-40                                         | Language          | german                                                   |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | only in the summer term                          | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1                                                | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 / 5,0                                          | Module owner      | Prof. Dr. Thomas Kürner                                  |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 150                                              |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 56                                               | Self studying (h) | 94                                                       |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | oral exam (20 min) or written exam (90 min)      |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Colloquium or lab journal                        |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                  |                   |                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                  |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                   |                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                  |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                  |                   |                                                          |
| Lecture notes<br>M. C. Jeruchim, P. Balaban, K. S. Shanmugan, Simulation of Communication Systems - Modeling, Methodology and Techniques, Kluwer 2000<br>R. Vaughan, J. B. Andersen, Channels, Propagation and Antennas for Mobile Communications, IEE Electromagnetic Waves Series 2003<br>J. G. Proakis, M. Saleh, Grundlagen der Kommunikationstechnik, Pearson Studium, 2. Auflage, 2004<br>M. Pätzold, Mobilfunkkanäle - Modellierung, Analyse und Simulation, Vieweg 1999<br>O. Beucher, MATLAB und Simulink, Pearson 2002<br>M. Schiff, Introduction to Communications Simulation, Artech House 2006<br>P. Stoica, R. Moses, Spectral Analysis of Signals, Pearson 2005 |                                                  |                   |                                                          |

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|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |

|                                                                                         |     |            |        |
|-----------------------------------------------------------------------------------------|-----|------------|--------|
|                                                                                         | 2,0 | Laboratory | german |
| <b>Literature</b>                                                                       |     |            |        |
| siehe Vorlesung                                                                         |     |            |        |
|                                                                                         | 2,0 | Lecture    | german |
| <b>Literature</b>                                                                       |     |            |        |
| Skript M.C.Jeruchim, P.Balaban, K.S.Shanmugan: Simulation of Communications - Modeling, |     |            |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                               |                   |                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------|-----------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Multimedia Databases                                                                          |                   |                                   |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4214610                                                                                       | Module version    | V2                                |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | INF-IS-61                                                                                     | Language          | german                            |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | irregular                                                                                     | Teaching unit     | Carl-Friedrich-Gauß-Fakultät      |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1 Semester                                                                                    | Institution       | Institut für Informations-systeme |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0 / 5,0                                                                                       | Module owner      | Prof. Dr. Wolf-Tilo Balke         |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 150                                                                                           |                   |                                   |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 42                                                                                            | Self studying (h) | 108                               |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                               |                   |                                   |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 graded work: Written exam, 90 minutes, or oral exam, approx. 30 minutes, or Take-Home-Exam. |                   |                                   |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1 non-graded work: 50% of the exercises must be passed                                        |                   |                                   |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                               |                   |                                   |
| <ul style="list-style-type: none"><li>• General structure of multimedia databases</li><li>• Advanced document types, multimedia documents</li><li>• Image-content search, low-level and high-level features</li><li>• High-dimensional indexing, inverted files, R-, M- and X-trees</li><li>• Search in audio files, acoustic features, e.g. pitch recognition</li><li>• Music retrieval, Hidden Markov Models, Query by Humming, etc.</li><li>• Video retrieval, segmentation and shot detection</li><li>• Video similarity, video signatures, abstracting and summaries</li></ul> |                                                                                               |                   |                                   |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                               |                   |                                   |
| Upon completion of this module, students will have basic knowledge in the field of multimedia databases.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                               |                   |                                   |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                               |                   |                                   |
| <ul style="list-style-type: none"><li>• Ingo Schmitt: Ähnlichkeitssuche in Multimedia-Datenbanken. Oldenbourg Wissenschaftsverlag, 2005.</li><li>• Vittorio Castelli, Lawrence D. Bergman: Image Databases. Wiley &amp; Sons, 2002.</li><li>• Ralf Steinmetz: Multimedia-Technologie: Grundlagen, Komponenten und Systeme. Springer Verlag, 1999.</li><li>• Setrag Khoshafian, Brad Baker: Multimedia and Imaging Databases. Morgan Kaufmann, 1996.</li></ul>                                                                                                                       |                                                                                               |                   |                                   |

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|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |

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|--|-----|------------------|----------------|
|  | 3,0 | Lecture/Exercise | english german |
|  | 1,0 | Exercise         | english        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                 |                   |                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Orientation Services Management                                                                                                                                                 |                   |                              |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2220180                                                                                                                                                                         | Module version    | V3                           |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | WW-AIP-18                                                                                                                                                                       | Language          | german                       |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | every term                                                                                                                                                                      | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                               | Institution       |                              |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 / 5,0                                                                                                                                                                         | Module owner      | Prof. Dr. David Woitschläger |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 150                                                                                                                                                                             |                   |                              |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 56                                                                                                                                                                              | Self studying (h) | 94                           |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                   |                              |
| Recommended requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Voraussetzung für das Modul sind Grundkenntnisse der Wirtschaftswissenschaften (Bachelor), beispielsweise des Dienstleistungsmanagement, des Marketing, der Unternehmensführung |                   |                              |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1 Klausur (120 min) oder 1 Take-at-Home-Exam                                                                                                                                    |                   |                              |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | für Organisation, Governance, Bildung / MA Sozialwissenschaften statt der Prüfungsleistung:<br>1 Klausur (120 min) oder 1 Take-at-Home-Exam                                     |                   |                              |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                 |                   |                              |
| „Ausgewählte Inhalte - abhängig von der Veranstaltungsauswahl:“ <ul style="list-style-type: none"><li>• Markenmanagement</li><li>• Gestaltung von Dienstleistungen</li><li>• Prozess- und Qualitätsmanagement</li><li>• Kundenwertorientiertes Beziehungsmanagement</li><li>• Customer Life-Cycle-Management</li><li>• Vertriebsmanagement</li><li>• Management von Dienstleistungsnetzwerken</li><li>• Methoden der Dienstleistungsforschung</li></ul>                                                                                         |                                                                                                                                                                                 |                   |                              |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                   |                              |
| Die Studierenden besitzen ein Verständnis über Fragestellungen, die sich im Rahmen der Gestaltung und Vermarktung von Dienstleistungen, dem Kundenbindungs-, Vertriebs- bzw. Markenmanagements stellen. Die Studierenden können auf Basis der erlernten Konzepte selbständig aktuelle betriebswirtschaftliche Fragestellungen in verschiedenen Branchenkontexten analysieren. Darüber hinaus verfügen sie über Methodenwissen zur qualitativen und quantitativen Analyse von Kunden- und Unternehmensdaten.                                     |                                                                                                                                                                                 |                   |                              |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                 |                   |                              |
| <ul style="list-style-type: none"><li>• Keller, Kevin L. (2008): Strategic Brand Management - Building, Measuring, and Managing Brand Equity, 3th ed., Prentice Hall.</li><li>• Johnston, Mark W. and Greg W. Marshall (2011): Sales Force Management, 10thed., McGraw-Hill.</li><li>• Kumar, V. and Werner Reinartz (2005): Customer Relationship Management: A Databased Approach, John Wiley &amp; Sons.</li><li>• Kumar, V. and Werner Reinartz (2012): Customer Relationship Management: Concept, Strategy, and Tools, Springer.</li></ul> |                                                                                                                                                                                 |                   |                              |





| <b>Related courses</b>                                                                                                                                                                                                                                                                                                                                          |     |                  |          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------|----------|
| <b>Rules for the choice of courses</b>                                                                                                                                                                                                                                                                                                                          |     |                  |          |
| Folgende Kombinationen sind wählbar: <ul style="list-style-type: none"> <li>• Variante A: Strategic Brand Management + Services Design/Business Model Innovation</li> <li>• Variante B: Customer Relationship Management + Sales Management/Vertriebsmanagement</li> </ul>                                                                                      |     |                  |          |
| <b>Compulsory attendance</b>                                                                                                                                                                                                                                                                                                                                    |     |                  |          |
|                                                                                                                                                                                                                                                                                                                                                                 |     |                  |          |
| Name of the course                                                                                                                                                                                                                                                                                                                                              | SWS | Eventtype        | Language |
| Strategic Brand Management                                                                                                                                                                                                                                                                                                                                      | 2,0 | Lecture          | german   |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                               |     |                  |          |
| <ul style="list-style-type: none"> <li>• Kevin L. Keller (2008): Strategic Brand Management</li> <li>• Ergänzende Literatur (PDF-Dokumente, Vorlesungsunterlagen zum Download)</li> </ul>                                                                                                                                                                       |     |                  |          |
| Services Design                                                                                                                                                                                                                                                                                                                                                 | 2,0 | Lecture          | english  |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                               |     |                  |          |
| <ul style="list-style-type: none"> <li>• Lehrbücher und weiterführende Literatur werden in den Veranstaltungen bekannt gegeben</li> <li>• Ergänzende Literatur (PDF-Dokumente, Vorlesungsunterlagen zum Download)</li> </ul>                                                                                                                                    |     |                  |          |
| Customer Relationship Management                                                                                                                                                                                                                                                                                                                                | 2,0 | Lecture          | german   |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                               |     |                  |          |
| <ul style="list-style-type: none"> <li>• #Lehrbücher und weiterführende Literatur werden in den Veranstaltungen bekannt gegeben</li> <li>• Ergänzende Literatur (PDF-Dokumente, Vorlesungsunterlagen zum Download)</li> </ul>                                                                                                                                   |     |                  |          |
| Sales Management                                                                                                                                                                                                                                                                                                                                                | 2,0 | Lecture          | german   |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                               |     |                  |          |
| <ul style="list-style-type: none"> <li>• Lehrbücher und weiterführende Literatur werden in den Veranstaltungen bekannt gegeben</li> <li>• Ergänzende Literatur (PDF-Dokumente, Vorlesungsunterlagen zum Download)</li> </ul>                                                                                                                                    |     |                  |          |
| Business Model Innovation: Concepts and Applications                                                                                                                                                                                                                                                                                                            | 4,0 | Lecture/Exercise | english  |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                               |     |                  |          |
| <ul style="list-style-type: none"> <li>• Lehrbücher und weiterführende Literatur werden in den Veranstaltungen bekannt gegeben</li> <li>• Ergänzende Literatur (PDF-Dokumente, Vorlesungsunterlagen zum Download)</li> </ul>                                                                                                                                    |     |                  |          |
| Customer Relationship Management and Customer Analytics                                                                                                                                                                                                                                                                                                         | 4,0 | Lecture/Exercise | german   |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                               |     |                  |          |
| <ul style="list-style-type: none"> <li>• Kumar, V. and Werner Reinartz (2018): Customer Relationship Management: Concept, Strategy, and Tools, 3. ed., Springer.</li> <li>• Lehrbücher und weiterführende Literatur werden in den Veranstaltungen bekannt gegeben</li> <li>• Ergänzende Literatur (PDF-Dokumente, Vorlesungsunterlagen zum Download)</li> </ul> |     |                  |          |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------|--------|
| Strategic Brand Management: Concepts and Applications                                                                                                                                                                                                                                                                                                                                                  | 4,0 | Lecture/Exercise | german |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                                      |     |                  |        |
| <ul style="list-style-type: none"><li>• Keller, Kevin L. und V. Swaminathan (2019): Strategic Brand Management - Building, Measuring, and Managing Brand Equity, 5th ed., Prentice Hall</li><li>• Ergänzende Journalpaper je Kapitel</li><li>• Vorlesungsunterlagen zum Download.</li><li>• Ergänzende Informationen und Literatur als Literaturverzeichnis und/oder zum Download zur Übung.</li></ul> |     |                  |        |

|                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                  |                   |                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                     | Orientation Marketing                                                                                                            |                   |                                       |
| Number                                                                                                                                                                                                                                                                                                                                                    | 2221110                                                                                                                          | Module version    | V2                                    |
| Shorttext                                                                                                                                                                                                                                                                                                                                                 | WW-MK-11                                                                                                                         | Language          | english german                        |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                        | only in the winter term                                                                                                          | Teaching unit     | Carl-Friedrich-Gauß-Fakultät          |
| Module duration                                                                                                                                                                                                                                                                                                                                           | 1                                                                                                                                | Institution       | Institut für Marketing und Innovation |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                     | 4 / 5,0                                                                                                                          | Module owner      | Prof. Dr. Wolfgang Fritz              |
| Workload (h)                                                                                                                                                                                                                                                                                                                                              | 150                                                                                                                              |                   |                                       |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                      | 56                                                                                                                               | Self studying (h) | 94                                    |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                  |                   |                                       |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                 | 1 exam (120 min) or 1 take-home exam                                                                                             |                   |                                       |
| Course achievement                                                                                                                                                                                                                                                                                                                                        | for Organisation, Governance, Education / MA Social Sciences instead of the examination:<br>1 exam (120 min) or 1 take-home exam |                   |                                       |
| Contents                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                  |                   |                                       |
| Selected content - depending on the event selection:<br><br>Special features of international marketing<br>Consumer behaviour and organisational buying behaviour<br>Techniques of data collection and data analysis in marketing                                                                                                                         |                                                                                                                                  |                   |                                       |
| Objective qualification                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                  |                   |                                       |
| The aim of the orientation module Marketing is to give students the opportunity to expand their knowledge in a subject that is not one of their specialisations. After completing the module, students will have a sound knowledge of the following areas: 1. buyer behaviour and marketing research, 2. international marketing                          |                                                                                                                                  |                   |                                       |
| Literature                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                  |                   |                                       |
| <ul style="list-style-type: none"><li>• Zentes, J./Swoboda, B./Schramm-Klein, H. (2006): Internationales Marketing, München 2006</li><li>• Kroeber-Riel, W./Weinberg, P./Gröppel-Klein, A. (2008): Konsumentenverhalten, 9. Aufl., München 2008</li><li>• Fantapié Altobelli, C. (2007): Marktforschung, Stuttgart 2007</li><li>• Folienskripte</li></ul> |                                                                                                                                  |                   |                                       |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |

|                                          |     |         |         |
|------------------------------------------|-----|---------|---------|
| Sustainability Transformation Management | 2,0 | Lecture | english |
| Consumer Behaviour                       | 2,0 | Lecture | german  |

|                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                  |                   |                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                     | Orientation Marketing                                                                                                            |                   |                                       |
| Number                                                                                                                                                                                                                                                                                                                                                    | 2221110                                                                                                                          | Module version    | V2                                    |
| Shorttext                                                                                                                                                                                                                                                                                                                                                 | WW-MK-11                                                                                                                         | Language          | english german                        |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                        | only in the winter term                                                                                                          | Teaching unit     | Carl-Friedrich-Gauß-Fakultät          |
| Module duration                                                                                                                                                                                                                                                                                                                                           | 1                                                                                                                                | Institution       | Institut für Marketing und Innovation |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                     | 4 / 5,0                                                                                                                          | Module owner      | Prof. Dr. Christof Backhaus           |
| Workload (h)                                                                                                                                                                                                                                                                                                                                              | 150                                                                                                                              |                   |                                       |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                      | 56                                                                                                                               | Self studying (h) | 94                                    |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                  |                   |                                       |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                 | 1 exam (120 min) or 1 take-home exam                                                                                             |                   |                                       |
| Course achievement                                                                                                                                                                                                                                                                                                                                        | for Organisation, Governance, Education / MA Social Sciences instead of the examination:<br>1 exam (120 min) or 1 take-home exam |                   |                                       |
| Contents                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                  |                   |                                       |
| Selected content - depending on the event selection:<br><br>Special features of international marketing<br>Consumer behaviour and organisational buying behaviour<br>Techniques of data collection and data analysis in marketing                                                                                                                         |                                                                                                                                  |                   |                                       |
| Objective qualification                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                  |                   |                                       |
| The aim of the orientation module Marketing is to give students the opportunity to expand their knowledge in a subject that is not one of their specialisations. After completing the module, students will have a sound knowledge of the following areas: 1. buyer behaviour and marketing research, 2. international marketing                          |                                                                                                                                  |                   |                                       |
| Literature                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                  |                   |                                       |
| <ul style="list-style-type: none"><li>• Zentes, J./Swoboda, B./Schramm-Klein, H. (2006): Internationales Marketing, München 2006</li><li>• Kroeber-Riel, W./Weinberg, P./Gröppel-Klein, A. (2008): Konsumentenverhalten, 9. Aufl., München 2008</li><li>• Fantapié Altobelli, C. (2007): Marktforschung, Stuttgart 2007</li><li>• Folienskripte</li></ul> |                                                                                                                                  |                   |                                       |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |

|                                                                                                                                                                                                                                                                                                        |     |          |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|--------|
| Consumer Behaviour                                                                                                                                                                                                                                                                                     | 2,0 | Lecture  | german |
| Sales Management                                                                                                                                                                                                                                                                                       | 2,0 | Lecture  | german |
| <b>Literature</b>                                                                                                                                                                                                                                                                                      |     |          |        |
| <ul style="list-style-type: none"><li>• Johnston, Mark W. und Marshall, Greg W. (2021). Sales Force Management – Leadership, Innovation, Technology, 13. ed., New York: Routledge.</li><li>• sowie zu den einzelnen Kapiteln weitere, in den Veranstaltungsunterlagen aufgeführte Literatur.</li></ul> |     |          |        |
| Sales Management                                                                                                                                                                                                                                                                                       | 2,0 | Exercise | german |
| <b>Literature</b>                                                                                                                                                                                                                                                                                      |     |          |        |
| <ul style="list-style-type: none"><li>• Johnston, Mark W. und Marshall, Greg W. (2021). Sales Force Management – Leadership, Innovation, Technology, 13. ed., New York: Routledge.</li><li>• sowie zu den einzelnen Kapiteln weitere, in den Veranstaltungsunterlagen aufgeführte Literatur.</li></ul> |     |          |        |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Orientation Information Management                                                                                                                                 |                   |                                                                             |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2222310                                                                                                                                                            | Module version    |                                                                             |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                    | Language          | german                                                                      |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | every term                                                                                                                                                         | Teaching unit     | Carl-Friedrich-Gauß-Fakultät                                                |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2                                                                                                                                                                  | Institution       | Institut für Wirtschafts-informatik - Abteilung Service-Informationssysteme |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 / 5,0                                                                                                                                                            | Module owner      | Prof. Dr. Susanne Robra-Bissantz                                            |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 150                                                                                                                                                                |                   |                                                                             |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 56                                                                                                                                                                 | Self studying (h) | 94                                                                          |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                    |                   |                                                                             |
| Recommended requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                             | The prerequisite for the module is basic knowledge in economics.                                                                                                   |                   |                                                                             |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1 assignment or 1 exam (120 min) or 1 portfolio or 1 take-at-home-exam or 1 exam+                                                                                  |                   |                                                                             |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | for Organisation, Governance, Education / MA Social Sciences instead of the examination:<br>1 term paper or 1 exam (120 min) or 1 portfolio or 1 take-at-home exam |                   |                                                                             |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                    |                   |                                                                             |
| Selected content – depending on the course selection: <ul style="list-style-type: none"><li>• Servicification</li><li>• Role of IT in Service Processes</li><li>• Service Dominant Logic</li><li>• Digital Products</li><li>• Value in Interaction</li><li>• E-Services</li><li>• Service Design</li><li>• Service Ecosystems and Platforms</li><li>• Digital Economy</li><li>• Digital Management</li><li>• Cooperation and Collaboration</li><li>• Digital Collaboration</li></ul> |                                                                                                                                                                    |                   |                                                                             |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                    |                   |                                                                             |
| Students understand the strategic relevance of information systems for companies in terms of operational tasks, people and technology. They are familiar with concepts for internal or inter-company IT-supported co-operation as well as their objectives and strategies in the context of strategic management. A possible specialisation is the view of application systems as e-services.                                                                                        |                                                                                                                                                                    |                   |                                                                             |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                    |                   |                                                                             |
| Course materials are provided on Stud.IP, along with additional literature.                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                    |                   |                                                                             |



| <b>Related courses</b>                                                                                                                                       |            |                  |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Rules for the choice of courses</b>                                                                                                                       |            |                  |                 |
| Colloquium optional                                                                                                                                          |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                                                 |            |                  |                 |
|                                                                                                                                                              |            |                  |                 |
| <b>Name of the course</b>                                                                                                                                    | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Digital Transformation: Collaborations                                                                                                                       | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                                                                                            |            |                  |                 |
| <ul style="list-style-type: none"> <li>• Vorlesungsunterlagen zum Download</li> <li>• Weitere Literatur wird in der Veranstaltung bekannt gegeben</li> </ul> |            |                  |                 |
| Digital Transformation: Services                                                                                                                             | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                                                                                            |            |                  |                 |
| Vorlesungsunterlagen per Download, weitere Literatur wird in der Veranstaltung bekannt gegeben.                                                              |            |                  |                 |
|                                                                                                                                                              | 2,0        | Colloquium       | german          |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Politics and Media                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |                                 |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1814730                                                                                                                                                                                                                                                                                                                                                                                                                                   | Module version    | V1                              |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | SW-MEW-73                                                                                                                                                                                                                                                                                                                                                                                                                                 | Language          | german                          |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | only in the winter term                                                                                                                                                                                                                                                                                                                                                                                                                   | Teaching unit     | Carl-Friedrich-Gauß-Fakultät    |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1 Semester                                                                                                                                                                                                                                                                                                                                                                                                                                | Institution       | Department Sozialwissenschaften |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4 / 9,0                                                                                                                                                                                                                                                                                                                                                                                                                                   | Module owner      | Prof. Dr. Nils C. Bandelow      |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 56                                                                                                                                                                                                                                                                                                                                                                                                                                        | Self studying (h) | 214                             |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 examination: <ul style="list-style-type: none"><li>• 1 written exam: 90 minutes or</li><li>• 1 oral examination: 20 minutes or</li><li>• 1 presentation and term paper (approx. 15 pages) on a special topic, the references to other topics of the module are to be explained or</li><li>• 1 portfolio or</li><li>• 1 project realisation with project report (approx. 15 pages)</li></ul> In each case as determined by the lecturer. |                   |                                 |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
| <ul style="list-style-type: none"><li>• Models and methods of political communication and policy analysis</li><li>• Case studies of political communication and policy field analysis, especially in the areas of health, welfare state, employment and education</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
| The political science module "Politics and Media" teaches skills that are in demand in fields such as political communication, for example in public relations, political consulting and public affairs. The module is also aimed at students who wish to acquire different perspectives for independent academic research. The academic foundation is laid by teaching selected theoretical and methodological approaches to political communication and policy field analysis. Students acquire the ability to apply these concepts to different contexts, particularly in the fields of media, health, education, labour and employment. The diverse interdisciplinary methodological approaches of policy field analysis are also analysed with reference to media and communication science concepts. Basically, the module imparts skills in understanding, presenting and categorising scientific results. |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |
| Will be announced in the courses and partly made available in Stud.IP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                                 |



| <b>Related courses</b>                                                                                                                    |            |                  |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Rules for the choice of courses</b>                                                                                                    |            |                  |                 |
| Seminar from upper division 1 (2 SWS) + seminar from upper division 2 (2 SWS) or seminar on both upper divisions (4 SWS) or block course. |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                              |            |                  |                 |
|                                                                                                                                           |            |                  |                 |
| <b>Name of the course</b>                                                                                                                 | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Social Conflicts and Public Affairs                                                                                                       | 4,0        | Seminar          | german          |
| Theories and case studies of political communication and policy field analysis.                                                           | 4,0        | Seminar          | german          |

|                                                                                                                                                                                                                                                                                                                                                                                             |                                                     |                   |                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                       | Planning of Terrestrial Radio Networks              |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                                      | 2424410                                             | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                   | ET-NT-41                                            | Language          | german                                                   |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                          | only in the summer term                             | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                             | 1                                                   | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                       | 4 / 5,0                                             | Module owner      | Prof. Dr. Thomas Kürner                                  |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                | 150                                                 |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                        | 56                                                  | Self studying (h) | 94                                                       |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                     |                                                     |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                   | Oral exam (20 minutes) or written exam (90 minutes) |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                          | Colloquium or lab journal                           |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                                    |                                                     |                   |                                                          |
| Introduction<br>Radio propagation models<br>Coverage planning<br>Planning cellular networks<br>General principles of cellular network planning<br>GSM radio network planning<br>UMTS radio network planning<br>Planning OFDMA networks<br><br>As part of the computer exercise, an introduction to the operation and handling of a radio planning tool                                      |                                                     |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                     |                                                     |                   |                                                          |
| After completing the module, students will have an understanding of the essential processes and interrelationships involved in planning terrestrial radio networks and will have acquired knowledge of the data required for this purpose and, in particular, the algorithms, models and methods used. They will be able to solve planning tasks independently using a radio planning tool. |                                                     |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                                  |                                                     |                   |                                                          |
| <ul style="list-style-type: none"><li>• Lecture notes (German and English)</li><li>• C. Lüders, Mobilfunksysteme, Vogel-Verlag 2001 #</li><li>• N. Geng, W. Wiesbeck, Planungsmethoden für die Mobilkommunikation, Springer-Verlag 1998</li><li>• J. Laiho, A. Wacker, T. Novosad, Radio Network Planning and Optimisation for UMTS, Wiley 2002</li></ul>                                   |                                                     |                   |                                                          |
| Remark                                                                                                                                                                                                                                                                                                                                                                                      |                                                     |                   |                                                          |
| This Master's module is also suitable for Bachelor students.                                                                                                                                                                                                                                                                                                                                |                                                     |                   |                                                          |



|                                                                                                                                                                                                                                                                           |            |                  |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                                                                                                                                                                                                    |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                                                                                                                                                                                                    |            |                  |                 |
|                                                                                                                                                                                                                                                                           |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                                                                                                                                                              |            |                  |                 |
|                                                                                                                                                                                                                                                                           |            |                  |                 |
| <b>Name of the course</b>                                                                                                                                                                                                                                                 | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                                                                                                                                                                                                                                                           | 2,0        | Laboratory       | german          |
| <b>Literature</b>                                                                                                                                                                                                                                                         |            |                  |                 |
| siehe Vorlesung                                                                                                                                                                                                                                                           |            |                  |                 |
|                                                                                                                                                                                                                                                                           | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                                                                                                                                                                                                         |            |                  |                 |
| Skript in deutscher und englischer Sprache C.Lüders, Mobilfunksysteme, Vogel-Verlag 2001 N.Geng, W.Wiesbeck, Planungsmethoden für die Mobilkommunikation, Springer-Verlag 1998 J.Laiho, A.Wacker, T.Novosad, Radio Network Planning and Optimisation for UMTS, Wiley 2002 |            |                  |                 |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                  | Computer Architecture II - Embedded Systems |                   |                                                          |
| Number                                                                                                                                                                 | 2416060                                     | Module version    |                                                          |
| Shorttext                                                                                                                                                              | ET-IDA-06                                   | Language          | german                                                   |
| Frequency of offer                                                                                                                                                     | only in the winter term                     | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                        | 1                                           | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                  | 4 / 6,0                                     | Module owner      | Prof. Dr. Selma Saidi                                    |
| Workload (h)                                                                                                                                                           | 180                                         |                   |                                                          |
| Class attendance (h)                                                                                                                                                   | 56                                          | Self studying (h) | 124                                                      |
| Compulsory requirements                                                                                                                                                |                                             |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                              |                                             |                   |                                                          |
| Course achievement                                                                                                                                                     |                                             |                   |                                                          |
| Contents                                                                                                                                                               |                                             |                   |                                                          |
|                                                                                                                                                                        |                                             |                   |                                                          |
| Objective qualification                                                                                                                                                |                                             |                   |                                                          |
|                                                                                                                                                                        |                                             |                   |                                                          |
| Literature                                                                                                                                                             |                                             |                   |                                                          |
| # Vorlesungsbegleitendes Material # W. Wolf, Computers As Components - Principles of Embedded Computing System Design, Morgan Kaufmann Publishers, ISBN 978-0123743978 |                                             |                   |                                                          |

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|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                        | 3,0        | Lecture          | german          |
|                                        | 1,0        | Exercise         | german          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                               |                   |                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                              | Specialisation Marketing                                                                                                                                                                                                                                      |                   |                                       |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2221120                                                                                                                                                                                                                                                       | Module version    | V3                                    |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                          | WW-MK-12                                                                                                                                                                                                                                                      | Language          | german                                |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                 | only in the summer term                                                                                                                                                                                                                                       | Teaching unit     | Carl-Friedrich-Gauß-Fakultät          |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                                                                                                                                                                                                                                                             | Institution       | Institut für Marketing und Innovation |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 / 5,0                                                                                                                                                                                                                                                       | Module owner      | Prof. Dr. Wolfgang Fritz              |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                       | 150                                                                                                                                                                                                                                                           |                   |                                       |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                               | 56                                                                                                                                                                                                                                                            | Self studying (h) | 94                                    |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                               |                   |                                       |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                          | 1 exam (60 min) or 1 take-at-home exam (2.5 CP)                                                                                                                                                                                                               |                   |                                       |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 written exam (60 min) or 1 exercise task or 1 take-at-home exam (for the exercise) (2.5 CP)<br><br>for Organisation, Governance, Education / MA Social Sciences instead of the examination additionally:<br>1 exam (60 min) or 1 take-at-home exam (2.5 CP) |                   |                                       |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                               |                   |                                       |
| Selected contents - depending on the course selection:<br><br>Selected aspects of distribution management<br>Techniques of data collection and data analysis in marketing<br>In-depth study of selected marketing topics using case studies and exercises                                                                                                                                                                                          |                                                                                                                                                                                                                                                               |                   |                                       |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                               |                   |                                       |
| After completing this module, students have a sound knowledge of the areas of distribution management, international marketing, buyer behaviour and marketing research. They are able to think through, structure and solve marketing problems of various kinds.                                                                                                                                                                                   |                                                                                                                                                                                                                                                               |                   |                                       |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                               |                   |                                       |
| <ul style="list-style-type: none"><li>• Zentes, J./Swoboda, B./Schramm-Klein, H. (2006): Internationales Marketing, München 2006</li><li>• Kroeber-Riel, W./Weinberg, P./Gröppel-Klein, A. (2008): Konsumentenverhalten, 9. Aufl., München 2008</li><li>• Fantapié Altobelli, C. (2007): Marktforschung, Stuttgart 2007</li><li>• Specht, G./Fritz, W. (2005): Distributionsmanagement, 4. Aufl., Stuttgart 2005</li><li>• Folienskripte</li></ul> |                                                                                                                                                                                                                                                               |                   |                                       |

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| <b>Related courses</b>                                                                                                                                                                                                                                                                                |            |                  |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Rules for the choice of courses</b>                                                                                                                                                                                                                                                                |            |                  |                 |
| Sales management and the exercise Sustainability Transformation Management are compulsory.<br>The exercise Sales Management is voluntary and only for exam preparation.<br>The order of the courses is arbitrary.                                                                                     |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                                                                                                                                                                                          |            |                  |                 |
|                                                                                                                                                                                                                                                                                                       |            |                  |                 |
| <b>Name of the course</b>                                                                                                                                                                                                                                                                             | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Sales Management                                                                                                                                                                                                                                                                                      | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                                                                                                                                                                                                                                     |            |                  |                 |
| <ul style="list-style-type: none"> <li>Johnston, Mark W. und Marshall, Greg W. (2021). Sales Force Management – Leadership, Innovation, Technology, 13. ed., New York: Routledge.</li> <li>sowie zu den einzelnen Kapiteln weitere, in den Veranstaltungsunterlagen aufgeführte Literatur.</li> </ul> |            |                  |                 |
| Exercise - Sustainability Transformation Management                                                                                                                                                                                                                                                   | 2,0        | Exercise         | english         |
| Sales Management                                                                                                                                                                                                                                                                                      | 2,0        | Exercise         | german          |
| <b>Literature</b>                                                                                                                                                                                                                                                                                     |            |                  |                 |
| <ul style="list-style-type: none"> <li>Johnston, Mark W. und Marshall, Greg W. (2021). Sales Force Management – Leadership, Innovation, Technology, 13. ed., New York: Routledge.</li> <li>sowie zu den einzelnen Kapiteln weitere, in den Veranstaltungsunterlagen aufgeführte Literatur.</li> </ul> |            |                  |                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                               |                   |                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                              | Specialisation Marketing                                                                                                                                                                                                                                      |                   |                                       |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2221120                                                                                                                                                                                                                                                       | Module version    | V4                                    |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                          | WW-MK-12                                                                                                                                                                                                                                                      | Language          | german                                |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                 | only in the summer term                                                                                                                                                                                                                                       | Teaching unit     | Carl-Friedrich-Gauß-Fakultät          |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                                                                                                                                                                                                                                                             | Institution       | Institut für Marketing und Innovation |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 / 5,0                                                                                                                                                                                                                                                       | Module owner      | Prof. Dr. Wolfgang Fritz              |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                       | 150                                                                                                                                                                                                                                                           |                   |                                       |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                               | 56                                                                                                                                                                                                                                                            | Self studying (h) | 94                                    |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                               |                   |                                       |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                          | 1 exam (60 min) or 1 take-at-home exam (2.5 CP)                                                                                                                                                                                                               |                   |                                       |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 written exam (60 min) or 1 exercise task or 1 take-at-home exam (for the exercise) (2.5 CP)<br><br>for Organisation, Governance, Education / MA Social Sciences instead of the examination additionally:<br>1 exam (60 min) or 1 take-at-home exam (2.5 CP) |                   |                                       |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                               |                   |                                       |
| Selected contents - depending on the course selection:<br><br>Selected aspects of distribution management<br>Techniques of data collection and data analysis in marketing<br>In-depth study of selected marketing topics using case studies and exercises                                                                                                                                                                                          |                                                                                                                                                                                                                                                               |                   |                                       |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                               |                   |                                       |
| After completing this module, students have a sound knowledge of the areas of distribution management, international marketing, buyer behaviour and marketing research. They are able to think through, structure and solve marketing problems of various kinds.                                                                                                                                                                                   |                                                                                                                                                                                                                                                               |                   |                                       |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                               |                   |                                       |
| <ul style="list-style-type: none"><li>• Zentes, J./Swoboda, B./Schramm-Klein, H. (2006): Internationales Marketing, München 2006</li><li>• Kroeber-Riel, W./Weinberg, P./Gröppel-Klein, A. (2008): Konsumentenverhalten, 9. Aufl., München 2008</li><li>• Fantapié Altobelli, C. (2007): Marktforschung, Stuttgart 2007</li><li>• Specht, G./Fritz, W. (2005): Distributionsmanagement, 4. Aufl., Stuttgart 2005</li><li>• Folienskripte</li></ul> |                                                                                                                                                                                                                                                               |                   |                                       |





|                                                                                   |            |                  |                 |
|-----------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                            |            |                  |                 |
| <b>Rules for the choice of courses</b>                                            |            |                  |                 |
| The lecture and exercise Sustainability Transformation Management are compulsory. |            |                  |                 |
| <b>Compulsory attendance</b>                                                      |            |                  |                 |
|                                                                                   |            |                  |                 |
| <b>Name of the course</b>                                                         | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Sustainability Transformation Management                                          | 2,0        | Lecture          | english         |
| Exercise - Sustainability Transformation Management                               | 2,0        | Exercise         | english         |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                      |                   |                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------|----------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Specialisation Information Management                                                |                   |                                  |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2222000010                                                                           | Module version    |                                  |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                      | Language          |                                  |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | every term                                                                           | Teaching unit     | Carl-Friedrich-Gauß-Fakultät     |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                                                                    | Institution       |                                  |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4 / 5,0                                                                              | Module owner      | Prof. Dr. Susanne Robra-Bissantz |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 150                                                                                  |                   |                                  |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 56                                                                                   | Self studying (h) | 94                               |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | The module “Orientation Service Information Systems” must be successfully completed. |                   |                                  |
| Recommended requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The prerequisite for the module is basic knowledge in economics.                     |                   |                                  |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                      |                   |                                  |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                      |                   |                                  |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                      |                   |                                  |
| Selected content which students apply in a practical project building on the knowledge gained in the “Orientation Service Information Systems”: <ul style="list-style-type: none"><li>• Servicification</li><li>• Role of IT in Service Processes</li><li>• Service Dominant Logic</li><li>• Digital Products</li><li>• Value in Interaction</li><li>• E-Services</li><li>• Service Design</li><li>• Service Ecosystems and Platforms</li><li>• Digital Economy</li><li>• Digital Management</li><li>• Cooperation and Collaboration</li><li>• Digital Collaboration</li></ul>                                                                                                                                                                                                                                                                              |                                                                                      |                   |                                  |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                      |                   |                                  |
| Students understand the strategic relevance of information systems for companies in terms of operational tasks, people and technology. They are familiar with concepts for internal and/or inter-company IT-supported co-operation as well as their objectives and strategies in the context of strategic management. A possible specialisation is the view of application systems as e-services. Students acquire the technical and methodological knowledge and skills to develop, design, critically reflect on, present and at least partially technically implement strategically relevant IT-supported innovations for companies. Through project work, they are familiarised with working in teams and with modern media and are therefore able to apply their knowledge, make it sustainably accessible for themselves and expand it independently. |                                                                                      |                   |                                  |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                      |                   |                                  |
| The course materials are available in Stud.IP; further reading can also be found there.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                      |                   |                                  |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                        | 4,0        | Project          | german          |
|                                        | 2,0        | Colloquium       | german          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                   |                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Spoken Language Processing                                                            |                   |                                                          |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2424680                                                                               | Module version    |                                                          |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ET-NT-68                                                                              | Language          |                                                          |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | only in the winter term                                                               | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                                                                                     | Institution       |                                                          |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 / 5,0                                                                               | Module owner      | Prof. Dr. Tim Fingscheidt                                |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 150                                                                                   |                   |                                                          |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 56                                                                                    | Self studying (h) | 94                                                       |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                       |                   |                                                          |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Oral exam 30 minutes or written exam 90 minutes (depending on number of participants) |                   |                                                          |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                       |                   |                                                          |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                       |                   |                                                          |
| <ul style="list-style-type: none"><li>- Basics of speech production and perception</li><li>- Feature extraction</li><li>- Hidden Markov models</li><li>- Acoustic models and language models</li><li>- Automatic speech recognition</li><li>- Spoken language systems</li></ul>                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                   |                                                          |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                       |                   |                                                          |
| After successful completion of the module, students will be able to classify time series (e.g., speech signals) using hidden Markov modeling. The students acquire all the necessary knowledge to suitably select, design, and evaluate methods and algorithms for automatic speech recognition to solve problems in practice.                                                                                                                                                                                                                                                                                                                            |                                                                                       |                   |                                                          |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                       |                   |                                                          |
| <ul style="list-style-type: none"><li>- Lecture slides</li><li>- X. Huang, A. Acero, H.-W. Hon: Spoken Language Processing, Prentice Hall, 2001</li><li>- B. Pfister, T. Kaufmann: Sprachverarbeitung, Springer, 2008</li><li>- A. Wendemuth: Grundlagen der Stochastischen Sprachverarbeitung, Oldenbourg, 2004</li><li>- E.G. Schukat-Talamazzini: Automatische Spracherkennung, Vieweg, 1995</li><li>- G.A. Fink: Mustererkennung mit Markov-Modellen, Teubner, 2003</li><li>- L. Rabiner, B.-H. Juang: Fundamentals of Speech Recognition, Prentice Hall, 1993</li><li>- K. Fukunaga: Statistical Pattern Recognition, Academic Press, 1990</li></ul> |                                                                                       |                   |                                                          |
| Remark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                       |                   |                                                          |
| This module from the master's program is also suitable for bachelor students. Basic knowledge of digital signal processing, as e.g. acquired in the module #digital signal processing#, facilitates the understanding of this lecture.                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                       |                   |                                                          |



| <b>Related courses</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                  |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Rules for the choice of courses</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |                 |
| <b>Name of the course</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2,0        | Lecture          | english         |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |                  |                 |
| <ul style="list-style-type: none"> <li>- X. Huang, A. Acero, H.-W. Hon: Spoken Language Processing, Prentice Hall, 2001</li> <li>- B. Pfister, T. Kaufmann: Sprachverarbeitung, Springer, 2008</li> <li>- A. Wendemuth: Grundlagen der Stochastischen Sprachverarbeitung, Oldenbourg, 2004</li> <li>- E.G. Schukat-Talamazzini: Automatische Spracherkennung, Vieweg, 1995</li> <li>- G.A. Fink: Mustererkennung mit Markov-Modellen, Teubner, 2003</li> <li>- L. Rabiner, B.-H. Juang: Fundamentals of Speech Recognition, Prentice Hall, 1993</li> <li>- K. Fukunaga: Statistical Pattern Recognition, Academic Press, 1990</li> </ul> |            |                  |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2,0        | Seminar          | english         |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |                  |                 |
| <ul style="list-style-type: none"> <li>- X. Huang, A. Acero, H.-W. Hon: Spoken Language Processing, Prentice Hall, 2001</li> <li>- B. Pfister, T. Kaufmann: Sprachverarbeitung, Springer, 2008</li> <li>- A. Wendemuth: Grundlagen der Stochastischen Sprachverarbeitung, Oldenbourg, 2004</li> <li>- E.G. Schukat-Talamazzini: Automatische Spracherkennung, Vieweg, 1995</li> <li>- G.A. Fink: Mustererkennung mit Markov-Modellen, Teubner, 2003</li> <li>- L. Rabiner, B.-H. Juang: Fundamentals of Speech Recognition, Prentice Hall, 1993</li> <li>- K. Fukunaga: Statistical Pattern Recognition, Academic Press, 1990</li> </ul> |            |                  |                 |

|                                                                                             |                         |                   |                                                          |
|---------------------------------------------------------------------------------------------|-------------------------|-------------------|----------------------------------------------------------|
| Title                                                                                       | Speech Communication    |                   |                                                          |
| Number                                                                                      | 2424500                 | Module version    | V1                                                       |
| Shorttext                                                                                   | ET-NT-50                | Language          |                                                          |
| Frequency of offer                                                                          | only in the winter term | Teaching unit     | Fakultät für Elektrotechnik, Informationstechnik, Physik |
| Module duration                                                                             | 1                       | Institution       |                                                          |
| Hours per Week / ECTS                                                                       | 4 / 5,0                 | Module owner      | Prof. Dr. Tim Fingscheidt                                |
| Workload (h)                                                                                | 150                     |                   |                                                          |
| Class attendance (h)                                                                        | 56                      | Self studying (h) | 94                                                       |
| Compulsory requirements                                                                     |                         |                   |                                                          |
| Expected performance/ Type of examination                                                   |                         |                   |                                                          |
| Course achievement                                                                          |                         |                   |                                                          |
| Contents                                                                                    |                         |                   |                                                          |
|                                                                                             |                         |                   |                                                          |
| Objective qualification                                                                     |                         |                   |                                                          |
|                                                                                             |                         |                   |                                                          |
| Literature                                                                                  |                         |                   |                                                          |
| - Kopien der Vorlesungsfolien - P.Vary u. R.Martin: Digital Speech Transmission, Wiley 2006 |                         |                   |                                                          |
| Remark                                                                                      |                         |                   |                                                          |
|                                                                                             |                         |                   |                                                          |



|                                                                                         |            |                  |                 |
|-----------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                  |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                  |            |                  |                 |
|                                                                                         |            |                  |                 |
| <b>Compulsory attendance</b>                                                            |            |                  |                 |
|                                                                                         |            |                  |                 |
| <b>Name of the course</b>                                                               | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                                                                         | 2,0        | Lecture          | german          |
| <b>Literature</b>                                                                       |            |                  |                 |
| Kopien der Vorlesungsfolien P.Vary u. R.Martin: Digital Speech Transmission, Wiley 2006 |            |                  |                 |

|                   |     |            |        |
|-------------------|-----|------------|--------|
|                   | 2,0 | Laboratory | german |
| <b>Literature</b> |     |            |        |
| siehe Vorlesung   |     |            |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                  |                   |                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Theory of Technology                                                                                                                                                                                                                                             |                   |                              |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2199300                                                                                                                                                                                                                                                          | Module version    |                              |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MW-STD-30                                                                                                                                                                                                                                                        | Language          | german                       |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | only in the summer term                                                                                                                                                                                                                                          | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1 Semester                                                                                                                                                                                                                                                       | Institution       |                              |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 6 / 9,0                                                                                                                                                                                                                                                          | Module owner      | Rolf F. Nohr                 |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                  |                   |                              |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 90                                                                                                                                                                                                                                                               | Self studying (h) | 180                          |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                  |                   |                              |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 Examination performance: <ul style="list-style-type: none"><li>• Term paper (12 to 15 pages) or</li><li>• Presentation with written paper (10 to 12 pages) or</li><li>• Written examination (120 minutes) or</li><li>• Oral examination (30 minutes)</li></ul> |                   |                              |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                  |                   |                              |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                  |                   |                              |
| The module aims to reflect on the development and technical design of media and media systems primarily from a media cultural studies, sociological, historical, philosophical and political perspective. Media development should not be understood here primarily as a linear process of innovation or the discovery of specific technologies. It is not distinct technical objects that represent the actual meaning-creating and meaning-producing core of the project of informatic, media or visual culture, but rather social structures of action on technologies. At the centre of the module are diverse opportunities to understand media technologies as large-scale technical systems and social practices and to examine their genesis and impact using tools such as technology theory, sociology of technology, historiography and discourse-theoretical analysis. |                                                                                                                                                                                                                                                                  |                   |                              |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                  |                   |                              |
| The module trains students to relate technologies and media effects. Historiographical development studies provide a better understanding of innovation discourses and the social significance of media. Explicitly interdisciplinary approaches expand the ability to recognise media objects and phenomena as contextually embedded. The ability to critically scrutinise the (alleged) factual should be deepened.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                  |                   |                              |
| The students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                  |                   |                              |
| <ul style="list-style-type: none"><li>• know, remember and understand current concepts, procedures and methods of technology theory and can combine them with media theoretical, historical and analytical approaches and apply them to innovation discourses, media objects and phenomena.</li><li>• can impart, acquire and critically evaluate knowledge discursively by means of presentation, discussion and argumentation, including cooperative forms of learning and group work.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                  |                   |                              |
| Specialist Competences: Students are able to theorise, historicise and analyse interdisciplinary media-technical and -cultural phenomena as well as to understand, classify, evaluate and critically reflect on current research questions and designs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                  |                   |                              |



Interdisciplinary Competences: Students are able to think and work in an interdisciplinary manner and to present, discuss, appropriate and critically evaluate complex theories and analyses, both alone and in collaborative groups.

#### Literature



#### Related courses

##### Rules for the choice of courses

- [1] Seminar (2 SWS)
- [2] Seminar (2 SWS)
- [3] Exercise (2 SWS)

##### Compulsory attendance

Work performance for the award of credits and grades:

Active participation in the courses: Lecture (lecturer presentation; preparation and follow-up, notes, self-study), seminar (literature study, presentation, discussion) and passing the examination. In order to achieve the qualification objectives for the seminars and exercises, it is necessary for the participants to take part in the interactive discussions and group work within the course. This requires the continuous presence of the participants, i.e. usually at least 80% of the attendance time.

| Name of the course                          | SWS | Eventtype | Language |
|---------------------------------------------|-----|-----------|----------|
| Master's Seminar in Technology Theory (HBK) |     | Seminar   | german   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                             |                   |                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | V4 Scientific and Technological Change                                                      |                   |                                                    |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4498430                                                                                     | Module version    |                                                    |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | GE-STD2-43                                                                                  | Language          |                                                    |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | every term                                                                                  | Teaching unit     | Fakultät für Geistes- und Erziehungswissenschaften |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                                                                                           | Institution       |                                                    |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 / 9,0                                                                                     | Module owner      | Prof. Dr. Bettina Wahrig                           |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Präsenzzeit: 60 h Selbststudium: 120 h Gesamtworkload: 270 h                                |                   |                                                    |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 60                                                                                          | Self studying (h) | 120                                                |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Die Basismodule B1 und B2 müssen vor dem Vertiefungsmodul V4 erfolgreich absolviert werden. |                   |                                                    |
| Recommended requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Erfolgreicher Abschluss der Module A1 und A2 sowie Aufbau TND1                              |                   |                                                    |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | selbstständige Hausarbeit (ca. 15-17 S. / ca. 30.000-34.000 Z.)                             |                   |                                                    |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                             |                   |                                                    |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                             |                   |                                                    |
| <p>- Z.B. technowissenschaftliche Artefakte, wissenschaftliche Praktiken, Methoden und Theorien im Kontext fiktionaler und non-fiktionaler Literatur, Epochenbegriffe, paradigmatische Entdeckungen und Erfindungen, das Experiment, das Labor, das Modell und der Prototyp in der Wissenschafts- und Technikphilosophie, Verhältnis von angewandter und Grundlagenforschung und deren Rolle für Entdeckungen und Erfindungen, Geschlechter- und Machtverhältnisse und ihr Einfluss auf die Konstitution und Dynamik von Erfindungen, Technologien und technisch-wissenschaftlicher Wandel.</p>                                                                                                                                                    |                                                                                             |                   |                                                    |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                             |                   |                                                    |
| <p>- Die Studierenden können die Erfindung und Wirkung grundlegender wissenschaftlicher Fakte und Artefakte kultur- und wissenschaftsgeschichtlich einordnen, in ihrer gesellschaftlichen Relevanz ermessen, ihre fiktionalen Spiegelungen auffinden, und historische Kontinuitäten und Diskontinuitäten im Hinblick auf aktuelle Hochtechnologien und Zukunftstechnologien interpretieren.</p> <p>- Die Studierenden können gesellschaftliche Probleme und Spannungsfelder, die an technowissenschaftliche Artefakte und Methoden geknüpft sind (etwa in den Bereichen Nukleartechnologie, Medizin- und Biotechnologie, Militärtechnologie, Luft- und Weltraumfahrt, Umweltwissen, Computer und Robotik) nach ethischen Kriterien beurteilen.</p> |                                                                                             |                   |                                                    |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                             |                   |                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                             |                   |                                                    |



|                                                                                            |            |                  |                 |
|--------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                     |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                     |            |                  |                 |
| 1 SE und eine weitere LV (frei zu wählen aus dem Lehrangebot des Moduls V4; WiSe und SoSe) |            |                  |                 |
| <b>Compulsory attendance</b>                                                               |            |                  |                 |
|                                                                                            |            |                  |                 |
| <b>Name of the course</b>                                                                  | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                                                                            |            | Seminar          |                 |

|                                             |                              |                   |                                  |
|---------------------------------------------|------------------------------|-------------------|----------------------------------|
| Title                                       | Scientific Working - Seminar |                   |                                  |
| Number                                      | 2218110                      | Module version    |                                  |
| Shorttext                                   | WW-WINFO-1                   | Language          | german                           |
| Frequency of offer                          | every term                   | Teaching unit     | Carl-Friedrich-Gauß-Fakultät     |
| Module duration                             | 1 Semester                   | Institution       |                                  |
| Hours per Week / ECTS                       | 0 / 4,0                      | Module owner      | Prof. Dr. Susanne Robra-Bissantz |
| Workload (h)                                |                              |                   |                                  |
| Class attendance (h)                        | 28                           | Self studying (h) | 92                               |
| Compulsory requirements                     |                              |                   |                                  |
| Expected performance/ Type of examination   |                              |                   |                                  |
| Course achievement                          |                              |                   |                                  |
| Contents                                    |                              |                   |                                  |
|                                             |                              |                   |                                  |
| Objective qualification                     |                              |                   |                                  |
|                                             |                              |                   |                                  |
| Literature                                  |                              |                   |                                  |
| abhängig von der konkreten Aufgabenstellung |                              |                   |                                  |



|                                            |            |                  |                 |
|--------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                     |            |                  |                 |
| <b>Rules for the choice of courses</b>     |            |                  |                 |
|                                            |            |                  |                 |
| <b>Compulsory attendance</b>               |            |                  |                 |
|                                            |            |                  |                 |
| <b>Name of the course</b>                  | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Master-Seminar Service Information Systems | 3,0        | Seminar          | german          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                        |                   |                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Science Communication in Practice                                                                                      |                   |                                 |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1818020                                                                                                                | Module version    |                                 |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | SW-MEW2-02                                                                                                             | Language          | german                          |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | every term                                                                                                             | Teaching unit     | Carl-Friedrich-Gauß-Fakultät    |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                        | Institution       | Department Sozialwissenschaften |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4 / 6,0                                                                                                                | Module owner      | Prof. Dr. Monika Tad-dicken     |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 180                                                                                                                    |                   |                                 |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 56                                                                                                                     | Self studying (h) | 124                             |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                        |                   |                                 |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 examination (in the interdisciplinary (MA OGB) or interdisciplinary area (MA Sowi) as coursework): Practical product |                   |                                 |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                        |                   |                                 |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                        |                   |                                 |
| In the exercise on science communication practice, various forms of internal and/or external science communication are presented and designed. The exercise includes the development, realisation and presentation of a concept for the publication of scientific findings (e.g. in the form of a radio report) and/or for communication within science. The exercise is carried out by lecturers with established knowledge of the practical field of science communication. A total of two different exercises must be completed. |                                                                                                                        |                   |                                 |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                        |                   |                                 |
| Students learn about the basic formats of applied science communication and create their own communication products in the form of texts, images or videos. They learn how different communication formats are created and acquire the relevant technical skills (e.g. handling a camera and microphone). This gives students a very good insight into the practical field of science communication and enables them to make their first contacts in the professional world through lecturers.                                      |                                                                                                                        |                   |                                 |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                        |                   |                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                        |                   |                                 |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
| Exercise + Exercise                    |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Campus on Air                          | 2,0        | Exercise         | german          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                    |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------|--------|
| Science in the AV-Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2,0 | Exercise           | german |
| <b>Literature</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |                    |        |
| <ul style="list-style-type: none"><li>• Sabine Streich, Videojournalismus. Ein Trainingshandbuch, Konstanz : UVK Verl.-Ges, 2008</li><li>• Anika Sehl, Günther Rager, Qualitätsmanagement im Videojournalismus. Eine qualitative Studie der ARD-Anstalten, Wiesbaden: Verlag für Sozialwissenschaften, 2008</li><li>• Matthias Walter, In Bewegung. Die Produktion von Web-Videos bei deutschen regionalen Tageszeitungen, Wiesbaden: Verlag für Sozialwissenschaften, 2010</li><li>• Martin Ordloff, Fernsehjournalismus, Konstanz: UVK Verl.-Ges., 2005</li></ul> |     |                    |        |
| Close to Science - in Dialogue with Science Journalists and PR Professionals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     | Lecture/Exercise   | german |
| Introduction to Practical Journalism                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2,0 | Practical exercise | german |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                      |                   |                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------|---------------------------------|
| Project Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                      |                   | 10 ECTS                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                      |                   |                                 |
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Project Module Communication Science |                   |                                 |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2199040                              | Module version    |                                 |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | SW-MEW2-04                           | Language          | german                          |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                      | Teaching unit     | Carl-Friedrich-Gauß-Fakultät    |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1 Semester                           | Institution       | Department Sozialwissenschaften |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0 / 10,0                             | Module owner      | Prof. Dr. Monika Tad-dicken     |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                      |                   |                                 |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2                                    | Self studying (h) | 298                             |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                      |                   |                                 |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 Examination: Portfolio             |                   |                                 |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                      |                   |                                 |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                      |                   |                                 |
| Completion, presentation and scientific documentation (portfolio) of a project in the field of communication sciences.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                      |                   |                                 |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                      |                   |                                 |
| Students are able to apply the knowledge, methodological competences and technical skills acquired during their studies in communication science project contexts.                                                                                                                                                                                                                                                                                                                                                                                                                |                                      |                   |                                 |
| Students are able to work independently on a scientific question in the field of communication and media studies. They master the working techniques required for the respective research project and can independently plan, carry out, analyse and document demanding research processes. Students have an overview of current research in a selected field of research and have mastered the relevant theoretical principles. They are able to present their research results competently and engage in a professional discussion.                                             |                                      |                   |                                 |
| In the project, students further develop their project management skills with a theoretical foundation. They expand their social skills such as teamwork, conflict management and time management, mediation skills in the application. They apply self-learning skills and take responsibility for the project goal with the help of increased self-management, self-organisation and independence. Motivation, reliability and resilience are tested in the field of application, as are problem-solving skills, creativity, process-orientation and project management skills. |                                      |                   |                                 |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |                   |                                 |
| Schenk, M.: Medienwirkungsforschung, 3. Aufl., Mohr Siebeck, Tübingen, 2007.<br>Further literature will be announced when the topics are assigned.                                                                                                                                                                                                                                                                                                                                                                                                                                |                                      |                   |                                 |



|                                                                                                                                   |            |                  |                 |
|-----------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                                                            |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                                                            |            |                  |                 |
| The module includes a one-semester independent project as well as documentation and reflection on the project (in the portfolio). |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                      |            |                  |                 |
|                                                                                                                                   |            |                  |                 |
| <b>Name of the course</b>                                                                                                         | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                                                                                                                   | 1,0        | Colloquium       | german          |
| <b>Literature</b>                                                                                                                 |            |                  |                 |
| Festlegung in den Sitzungen.                                                                                                      |            |                  |                 |



|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                        |                   |                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------|------------------------------|
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                        | Research Project in Media Technology                   |                   |                              |
| Number                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1814120                                                | Module version    |                              |
| Shorttext                                                                                                                                                                                                                                                                                                                                                                                                                                    | SW-MEW-12                                              | Language          | german                       |
| Frequency of offer                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                        | Teaching unit     | Carl-Friedrich-Gauß-Fakultät |
| Module duration                                                                                                                                                                                                                                                                                                                                                                                                                              | 1 Semester                                             | Institution       |                              |
| Hours per Week / ECTS                                                                                                                                                                                                                                                                                                                                                                                                                        | 0 / 10,0                                               | Module owner      | Prof. Dr. Eduard Jorswieck   |
| Workload (h)                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                        |                   |                              |
| Class attendance (h)                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                                                      | Self studying (h) | 298                          |
| Compulsory requirements                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                        |                   |                              |
| Expected performance/ Type of examination                                                                                                                                                                                                                                                                                                                                                                                                    | 1 Examination: Project presentation and documentation. |                   |                              |
| Course achievement                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                        |                   |                              |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                        |                   |                              |
| The project task is defined from the research area of electronic media from the current project environment of the lecturer offering the course. The teaching content depends on the respective task.                                                                                                                                                                                                                                        |                                                        |                   |                              |
| Objective qualification                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                        |                   |                              |
| Students carry out a completed, practical task related to a current research project at the Institute of Communications Engineering and thus acquire key qualifications, such as the ability to plan and coordinate a project independently, to divide up the task and to define and adhere to milestones. Knowledge and skills in the field of media technology acquired during the degree programme are applied in practice and presented. |                                                        |                   |                              |
| Students acquire project-related or professional field-related skills, such as the ability to work together across disciplines and social skills such as teamwork, conflict resolution, project management skills and mediation skills.                                                                                                                                                                                                      |                                                        |                   |                              |
| Literature                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                        |                   |                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                        |                   |                              |



|                                        |            |                  |                 |
|----------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                 |            |                  |                 |
| <b>Rules for the choice of courses</b> |            |                  |                 |
|                                        |            |                  |                 |
| <b>Compulsory attendance</b>           |            |                  |                 |
|                                        |            |                  |                 |
| <b>Name of the course</b>              | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
|                                        | 1,0        | Project          | german          |

|                                                                                                                                                                                                                                                       |                                                                                                                                                                 |                   |                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------|
| Master's Thesis                                                                                                                                                                                                                                       |                                                                                                                                                                 |                   | 30 ECTS                         |
|                                                                                                                                                                                                                                                       |                                                                                                                                                                 |                   |                                 |
| Title                                                                                                                                                                                                                                                 | Master's Thesis                                                                                                                                                 |                   |                                 |
| Number                                                                                                                                                                                                                                                | 18118060                                                                                                                                                        | Module version    |                                 |
| Shorttext                                                                                                                                                                                                                                             | SW-MEW2-06                                                                                                                                                      | Language          |                                 |
| Frequency of offer                                                                                                                                                                                                                                    | every term                                                                                                                                                      | Teaching unit     | Carl-Friedrich-Gauß-Fakultät    |
| Module duration                                                                                                                                                                                                                                       | 1 Semester                                                                                                                                                      | Institution       | Department Sozialwissenschaften |
| Hours per Week / ECTS                                                                                                                                                                                                                                 | 1 / 30,0                                                                                                                                                        | Module owner      | Prof. Dr. Monika Tad-dicken     |
| Workload (h)                                                                                                                                                                                                                                          |                                                                                                                                                                 |                   |                                 |
| Class attendance (h)                                                                                                                                                                                                                                  | 14                                                                                                                                                              | Self studying (h) | 886                             |
| Compulsory requirements                                                                                                                                                                                                                               | You can register for the Master's thesis if you have 72 credit points. The degree profile (M.A. or M.Sc.) is stated in the application for the Master's thesis. |                   |                                 |
| Expected performance/ Type of examination                                                                                                                                                                                                             | 1 Examination: Six-month Master's thesis                                                                                                                        |                   |                                 |
| Course achievement                                                                                                                                                                                                                                    |                                                                                                                                                                 |                   |                                 |
| Contents                                                                                                                                                                                                                                              |                                                                                                                                                                 |                   |                                 |
| The content depends on the specific task. The Master's thesis is discussed and presented in a Master's colloquium.                                                                                                                                    |                                                                                                                                                                 |                   |                                 |
| Objective qualification                                                                                                                                                                                                                               |                                                                                                                                                                 |                   |                                 |
| On completion of the module, students demonstrate that they have independently worked on a topic in communication science or media technology or an interdisciplinary topic with a focus on one of the two study profiles with a scientific approach. |                                                                                                                                                                 |                   |                                 |
| Students are able to develop a topic independently, approach it by means of a suitable question and conceptualise and work on it both theoretically and methodically.                                                                                 |                                                                                                                                                                 |                   |                                 |
| They deepen key academic skills by applying working techniques for researching and analysing academic literature as well as the linguistic and formal design of an academic paper.                                                                    |                                                                                                                                                                 |                   |                                 |
| Students are able to present the main results of their work in a targeted manner and evaluate the results.                                                                                                                                            |                                                                                                                                                                 |                   |                                 |
| Literature                                                                                                                                                                                                                                            |                                                                                                                                                                 |                   |                                 |
| Depending on the chosen Master's topic                                                                                                                                                                                                                |                                                                                                                                                                 |                   |                                 |



|                                                                                                                                                         |            |                  |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|
| <b>Related courses</b>                                                                                                                                  |            |                  |                 |
| <b>Rules for the choice of courses</b>                                                                                                                  |            |                  |                 |
| The Master's thesis can be supervised by any of the institutes involved in the Master's degree program in Digital Communication and Media Technologies. |            |                  |                 |
| <b>Compulsory attendance</b>                                                                                                                            |            |                  |                 |
|                                                                                                                                                         |            |                  |                 |
| <b>Name of the course</b>                                                                                                                               | <b>SWS</b> | <b>Eventtype</b> | <b>Language</b> |
| Exam Colloquium in Communication Science                                                                                                                | 1,0        | Colloquium       | german          |

